

PLDP-IDS: Personalized Local Differential Privacy for Infinite Data Stream

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ABSTRACT

Streaming data collection is widely used for real-time analytics, such as event monitoring, where user data is continuously reported to a collector. Protecting privacy in this setting is challenging, as simple data publication may expose sensitive information. Local Differential Privacy (LDP) has therefore become a standard technique for safeguarding individuals while preserving data utility. However, most existing LDP studies on data streams assume a uniform privacy level across users or time slots, which often leads to suboptimal utility. In this paper, we address this limitation by proposing two baseline population-division solutions, PLPD and PLPA, which tailor privacy protection to personalized requirements by allocating and recycling users across time slots, while maintaining accurate stream estimation. Building on these baselines, we further develop two compatible enhancement modules—*Optimal Population Selection* (OPS) to reduce sampling error, and *Personalized Utility Enhancement* (PUE) to improve the accuracy of highly protected data by leveraging lower-privacy reports. Integrating these modules yields two enhanced methods, PLPD+ and PLPA+, which achieve significantly improved utility while preserving personalized privacy guarantees. We validate the efficiency and effectiveness of our four methods by conducting extensive experiments on both real-world and synthetic datasets. Experimental results demonstrate that our approaches achieve significantly higher accuracy than traditional non-personalized methods, reducing the average error by 43.18% on real datasets and 41.88% on synthetic datasets.

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/dulei715/WEeventPLDPCode>.

1 INTRODUCTION

With the widespread adoption of smart devices and high-quality wireless networks, users continuously generate large volumes of streaming data that are collected and analyzed in real time to support various online services, such as traffic monitoring and location-based recommendation. However, directly collecting and publishing streaming data lead to serious privacy leakage, discouraging user participation in many real-world applications. For example, an HIV-positive person may refuse to participate in a related investigation due to privacy concerns [15]. As a result, providing strong privacy protection while maintaining high data utility has become a fundamental challenge in streaming data analysis.

Local Differential Privacy (LDP) [36] is a trust-free privacy model in which users perturb their data locally before reporting, enabling decentralized data collection with formal privacy guarantees and practical utility. Accordingly, LDP has been widely adopted in real-world deployments. In streaming applications, however, user reports are often temporally correlated. To capture such correlations, w -event LDP [32] has been introduced to protect sequences of related events within a sliding window of length w , thereby supporting continual data collection over infinite streams.

However, existing w -event LDP mechanisms typically assume a uniform window size and single privacy budget shared by all users. In practice, users often exhibit heterogeneous temporal patterns and diverse privacy preferences. For example some users (e.g., entertainers) consider their location highly sensitive and thus require protection over longer correlated event blocks (i.e., a large w), whereas others (e.g., street artists) may be more willing to disclose their locations and prefer shorter windows (i.e., a smaller w) for

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higher utility or visibility. As a result, enforcing a global window size and privacy budget can overprotect many users and substantially degrade estimation accuracy. These observations motivate personalized w -event privacy, where each user specifies an individual window size and privacy budget that reflect their temporal sensitivity and privacy requirements. We illustrate this setting with a privacy-preserving online car-hailing example in Figure 1.

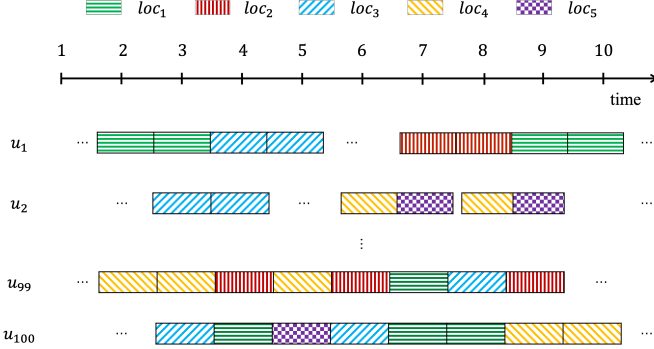


Figure 1: An example for personalized w -event

Example 1. Consider $n = 100$ drivers $U = \{u_1, \dots, u_{100}\}$ moving among $d = 5$ locations, $\{loc_1, \dots, loc_5\}$ (Figure 1). The maximum related-event block size is 4 for $u_1 - u_{98}$ and 8 for u_{99}, u_{100} . Under traditional uniform w -event privacy, we need to set $w = 8$. With total privacy budget $\epsilon = 1$, the standard error is:

$$SE = \sqrt{\frac{\text{Var}_{FO} \left[\frac{\epsilon}{w}, n \right]}{n}} = \sqrt{\frac{d - 2 + e^{\epsilon/w}}{(e^{\epsilon/w} - 1)^2}} \approx 15.27.$$

However, assigning $w = 8$ to all users overprotect $u_1 - u_{98}$. If we instead partition users into two groups and apply $w_1 = 4$ for the former $n_1 = 98$ users and $w_2 = 8$ for the latter $n_2 = 2$ users, the error becomes:

$$SE = \sqrt{\frac{\text{Var}_{FO} \left[\frac{\epsilon}{w_1}, n_1 \right] + \text{Var}_{FO} \left[\frac{\epsilon}{w_2}, n_2 \right]}{n}} \approx 7.53,$$

which is substantially lower than the uniform 8-event setting.

Personalized w -event privacy has recently been studied for infinite data streams under the Centralized Differential Privacy (CDP) model [10], where a trusted curator centrally collects data and injects noise to ensure privacy. While this line of work demonstrates that allowing heterogeneous window sizes and privacy budgets can significantly improve utility, it fundamentally relies on centralized access to data and global coordination across users. In contrast, we consider personalized w -event privacy under the LDP model, where each user perturbs his/her data locally and no trusted third party exists. This shift in the privacy model introduces *new challenges* that cannot be addressed by directly extending centralized solutions [10], including how to *manage population resources*, *determine effective sampling strategies*, and *optimize utility* under heterogeneous privacy requirements with only locally perturbed reports.

Specifically, we conduct a detailed analysis of error characteristics introduced by Personalized Local Differential Privacy (PLDP) [17] mechanisms in streaming data collection. Based on the theoretical foundation of PLDP, we propose two baseline methods, PLDP

Population Distribution (PLPD) and PLDP Population Absorption (PLPA), to improve data utility under personalized privacy settings. However, these two methods do not fully exploit the heterogeneity of personalized window sizes, nor do they leverage the information published under lower privacy protection (i.e., higher utility) to enhance the estimation of high-privacy data. To overcome these limitations, we further design PLDP Population Distribution Plus (PLPD⁺) and PLDP Population Absorption Plus (PLPA⁺), which choose the best window size with the lowest error and improve the estimation of high-privacy data using low-privacy data, achieving superior overall utility.

Contributions. We summarize our contributions as follows:

- We formally define the problem of Personalized Private Streaming Data Estimation in Local Setting, which focuses on enable statistical analysis of personalized data streams while ensuring personalized privacy protection (i.e., w -event PLDP) in Section 3.
- We propose two effective population division-based methods PLDP Population Distribution (PLPD) and PLDP Population Absorption (PLPA) under PLDP in Section 4.
- We further introduce two enhanced methods PLDP Population Distribution Plus (PLPD⁺) and PLDP Population Absorption Plus (PLPA⁺) to improve the utility in Section 5.
- We evaluate the proposed methods on both real and synthetic datasets to demonstrate their efficiency and effectiveness in Section 6.

2 RELATED WORK

2.1 Local Differential Privacy on Streaming Data

Local Differential Privacy (LDP)[3] enables privacy-preserving data collection without a trusted curator and has been widely adopted in practice. Existing studies on LDP for streaming data can be broadly categorized into event-level, user-level, and w -event privacy models.

Event-level LDP (event-LDP) mechanisms, such as RAPPOR [14] and ToPL [37], perturb each record independently and do not account for temporal correlations. User-level LDP methods [2, 27, 40, 41] aim to protect entire user streams, but typically suffer from rapid privacy budget consumption or limited adaptability in infinite streams. To bridge the gap between event-level and user-level protection, w -event privacy was introduced to protect event sequences within a sliding window of length w [19]. Ren et al. [32] extend this notion to the local setting and propose LDP-IDS for infinite data streams. However, LDP-IDS assumes a uniform window size and privacy budget for all users, which limits its applicability in personalized settings. In contrast, we study personalized w -event privacy under LDP.

2.2 Personalized Differential Privacy

Personalized Differential Privacy (PDP), also known as Heterogeneous Differential Privacy (HDP) [1], allows heterogeneous privacy requirements across users or data attributes. Existing PDP models can be broadly divided into item-grained and user-grained settings.

Table 1: Summary for related work.

Categories	Model Types	Methods	Infinite & correlated	Trust-free	Personalized
LDP on streaming data	event-level LDP	RAPPOR [14]	×	✓	×
		ToPL [37]	×	✓	×
	user-level LDP	CGM [2]	✓	✓	×
		DDRM [40]	✓	✓	×
		StaSwitch [41]	✓	✓	×
		PrivShape [27]	✓	✓	×
	w-event LDP	LDP-IDS [32]	✓	✓	×
Personalized DP	item-grained privacy	OSDP [22]	×	×	✓
		Pufferfish [9, 21, 29–31]	✓	×	✓
		Blowfish [13, 16, 25]	✓	×	✓
		PRR [35]	×	✓	✓
	user-grained privacy	SM [18]	×	×	✓
		AdaPDP [28]	×	×	✓
		ADPM [6]	×	×	✓
		PCE [7]	×	✓	✓
		HDG-COE [23]	×	✓	✓
		PLDP-MRQ [17]	×	✓	✓
Out methods			✓	✓	✓

Item-grained Privacy. Item-grained approaches, such as One-Sided Differential Privacy (OSDP) [22], Pufferfish [21] and Blowfish [16], relax protection for less sensitive attributes to improve utility. However, they rely on centralized mechanisms and policy specification, making them unsuitable for decentralized streaming settings.

User-grained Privacy. User-grained PDP allows each user to specify an individual privacy budget. Representative works include Differential Privacy (PDP) [18], Adaptive Personalized Differential Privacy (AdaPDP) [28], and Affine Differentially Private Mean Estimator (ADPM) [6], which primarily focus on centralized or static data. More recently, Chen et al. [7] and He et al. [17] propose Personalized Local Differential Privacy (PLDP) frameworks for count and range queries. Li et al. [23] propose HDG-COE, which enables users to specify personalized indistinguishable areas for protection, achieving high utility. However, existing PLDP methods do not consider temporal correlations in infinite data streams.

Beyond LDP and PLDP, several studies investigate streaming data privacy under centralized DP with a trusted curator, including w-event privacy [19], ℓ -trajectory privacy [4], and (w, ϵ) -personalized privacy [10]. In contrast, we study personalized local differential privacy for infinite, correlated attribute streams without requiring any trusted curator. Table 1 summarizes representative studies.

3 PROBLEM SETTINGS

In this section, we first introduce key conceptions related to data stream. Next, we present our new privacy definition: w -Event ϵ -Personalized Local Differential Privacy ((w, ϵ) -EPLDP) based on the conception of Personalized Local Differential Privacy (PDP). Finally, we provide the problem definition: Personalized Private Steaming Data Estimation in Local Setting (PPSDELS). Table 2 summarizes the notions used in this paper.

3.1 Data Streams

We follow the standard streaming data model [5, 19]. At each time slot t , users report attribute values from a finite domain Ω of size d , forming a database D_t . The sequence $S = \langle D_1, D_2, \dots \rangle$ constitutes an infinite data stream. The server aims to estimate the frequency distribution f_t at each time slot under local privacy constraints.

Table 2: Notations.

Notations	Description
u_i	the i -th user
n	the number of users in U
w	the privacy window size of U
w_i	u_i 's privacy window size
ϵ	the privacy budget set of U
ϵ_i	u_i 's privacy budget
$\tilde{\epsilon}$	the unique list of ϵ
$\tilde{\epsilon}_k$	the k -th value in $\tilde{\epsilon}$
g	the frequency list of users' privacy budget requirement
g_k	the frequency of users requiring $\tilde{\epsilon}_k$ (i.e., $g_k = g[k]$)
d	the domain size of the attribute
m	the number of unique privacy budget requirements
f	the real statistical histogram at any time slot
f_t	the real statistical histogram at time slot t
$\hat{f}_t$	the estimation statistical histogram at time slot t
h	the obfuscated statistical frequency at any time slot
h_t	the obfuscated statistical frequency at time slot t
ω_j	the j -th value in the value domain
$f_j, \hat{f}[j]$	the frequency that ω_j occurs
$f_j, \hat{f}[j]$	the estimation value of f_j
$h_j, h[j]$	the obfuscated frequency that ω_j occurs

Definition 1 (w -Neighboring Stream Prefixes [5, 19]). Let w be a positive integer. Two stream prefixes S_t, S'_t are w -neighboring (i.e., $S_t \sim_w S'_t$), if:

- (1) for each $S_t[\tau], S'_t[\tau]$ such that $\tau \leq t$ and $S_t[\tau] \neq S'_t[\tau]$, it holds that $S_t[\tau]$ and $S'_t[\tau]$ are neighboring [19] in centralized DP, and
- (2) for each $S_t[\tau_1], S_t[\tau_2], S'_t[\tau_1], S'_t[\tau_2]$ with $\tau_1 \leq \tau_2$, $S_t[\tau_1] \neq S'_t[\tau_1]$ and $S_t[\tau_2] \neq S'_t[\tau_2]$, it holds that $\tau_2 - \tau_1 + 1 \leq w$.

3.2 Privacy Definitions

Let $\mathcal{U}$ be the domain of users and $\mathcal{E}$ be the domain of privacy budgets. Let $U \subseteq \mathcal{U}$ be the user set with n users, specifically, $U = \{u_1, \dots, u_n\}$. Let $\epsilon : \mathcal{U} \rightarrow \mathcal{E}$ be the privacy budget requirement function, where $\epsilon(u_i) = \epsilon_i$ indicates that u_i requires ϵ_i privacy budget. We define $\tilde{\epsilon} = \text{Unique}(\epsilon)$ as the unique privacy budget list with size $m = |\tilde{\epsilon}|$. Let $\hat{\mathcal{D}}_k$ be the database domain protected by $\tilde{\epsilon}_k$ with $k \in [m]$.

Definition 2 (Budget Group and Budget Group Data). Given a unique privacy budget list $\tilde{\epsilon}$ with $|\tilde{\epsilon}| = m$ and a user set $U \subseteq \mathcal{U}$ with n users, where each user u_i holds a privacy budget $\epsilon_i \in \tilde{\epsilon}$, U can be divided into m groups $\{U_1, U_2, \dots, U_m\}$ based on distinct privacy budgets. The pair $BG_j = (\epsilon_j, U_j)$ is called the j -th Budget Group of U . Let Y_j be the output of each $u_i \in U_j$ applying ϵ_j -Local Differential Privacy mechanism on the data domain Ω . The tuple $BGD_j = (\epsilon_j, U_j, Y_j)$ is called the j -th Budget Group Data of U .

Definition 3 (ϵ -Personalized Local Differential Privacy, ϵ -PLDP [17]). A mechanism $\mathcal{M}$ satisfies ϵ -PLDP if and only if, for any user u with possible input values $x, x' \in \hat{\mathcal{D}}_k \subseteq \hat{\mathcal{D}}_k$,

$$\forall y \in \text{Range}(\mathcal{M}) : \Pr[\mathcal{M}(x) = y] \leq e^{\epsilon(u)} \Pr[\mathcal{M}(x') = y],$$

where $\text{Range}(\mathcal{M})$ denotes the set of all possible outputs of $\mathcal{M}$.

In traditional personalized local differential privacy (PLDP), each user is associated with a single privacy parameter ϵ_i , which controls the perturbation strength of individual data reports. However, in

Table 3: Privacy Level Table

$\epsilon \backslash w$	1.5	1.0	0.5
10	Level (1, 1)	Level (1, 2)	Level (1, 3)
20	Level (2, 1)	Level (2, 2)	Level (2, 3)
30	Level (3, 1)	Level (3, 2)	Level (3, 3)

streaming data settings, each user requires two parameters: w_i , which specifies the event window size and bounds the *temporal scope* of privacy protection, and ϵ_i , which determines the *strength* of per-event perturbation. Importantly, these two parameters capture *orthogonal dimensions* of privacy risk and are not interchangeable. While ϵ_i controls the indistinguishability of individual events, w_i constrains the amount of temporal correlation that an adversary can exploit across consecutive reports. As a result, two mechanisms with identical overall privacy budgets but different window sizes can exhibit substantially different privacy leakage in the presence of correlated data streams. This distinction is especially critical in personalized settings, where users may share similar ϵ_i but require fundamentally different temporal protections.

Definition 4 (w -Event ϵ -Personalized Local Differential Privacy, (w, ϵ) -EPLDP). Let $\mathcal{M}$ be a mechanism that takes a stream prefix of arbitrary τ size as inputs. Let $\mathcal{O}$ be the set of all possible outputs of $\mathcal{M}$. For any universe set of users $U = \{u_1, u_2, \dots, u_{|U|}\}$, $\mathcal{M}$ satisfies (w, ϵ) -EPLDP if $\forall w_i \in w, \forall S_\tau, S'_\tau$ satisfying $S_\tau \sim_{w_i} S'_\tau$ and $\forall O \subseteq \mathcal{O}$, it holds that

$$\Pr[\mathcal{M}(S_\tau) \in O] \leq e^{\epsilon(u_i)} \Pr[\mathcal{M}(S'_\tau) \in O],$$

where u_i requires w_i -event ϵ_i -LDP privacy protection. When the output set is discrete, i.e. $O = \{y_1, y_2, \dots, y_{|O|}\}$, the condition above can also be rewritten as

$$\Pr[\mathcal{M}(S_\tau) = y] \leq e^{\epsilon(u_i)} \Pr[\mathcal{M}(S'_\tau) = y].$$

Similar to the formulation in CDP setting [10], the pair (w_i, ϵ_i) is denoted as u_i 's *privacy requirement* [10]. In particular, when $w_i \equiv 1$, (w, ϵ) -EPLDP collapses to ϵ -PLDP [17]. Additionally, when (w_i, ϵ_i) is a constant pair (i.e., (w, ϵ)), it collapses to w -Event ϵ -LDP [19, 32].

In practice, users can specify their own privacy requirements (w_i, ϵ_i) based on their privacy preferences and system-assigned values. Assume the value domains of w_i and ϵ_i are $\{10, 20, 30\}$ and $\{0.5, 1.0, 1.5\}$, respectively. This results in 9 possible privacy levels, as shown in Table 3, where a higher value indicates stronger privacy protection. The system provides all users with these 9 privacy levels: Level (1, 1), Level (1, 2), ..., Level (3, 3). Users select one of these levels according to their needs. After receiving a level (e.g., Level (2, 3)), the system translates it into specific privacy parameters (e.g., (20, 0.5)). This design simplifies the process for users to choose privacy settings while preventing malicious manipulation of privacy parameters.

3.3 Problem Definition

Based on the above definitions, the server observes only locally perturbed reports and reconstructs an estimated frequency stream $\langle \hat{f}_1, \hat{f}_2, \dots \rangle$. The goal is to minimize estimation error while satisfying (w, ϵ) -EPLDP.

Definition 5. (PPSDELS Problem). Given any positive integer $T \in \mathbb{N}^+$, a user set $U = \{u_1, u_2, \dots, u_n\} \in \mathcal{U}$, where each u_i holds a privacy requirement pair (w_i, ϵ_i) and a series of data $x_{i,t}$ for $t \in [T]$, all the $x_{i,t}$ at time slot t form D_t , and all the D_t form a data stream prefix $S_T = \langle D_1, D_2, \dots, D_T \rangle$. PPSDELS is to publish an obfuscated stream prefix frequency estimation $\langle \hat{f}_1, \hat{f}_2, \dots, \hat{f}_T \rangle$ of S_T achieving (w, ϵ) -EPLDP with the error between $\hat{f}$ and f minimized, namely:

$$\min \sum_{t \in [T]} \|\hat{f}_t - f_t\|_2^2$$

$$s.t. \quad \sum_{\tau=\min(t-w_i+1, 1)}^t \epsilon_{i,\tau} \leq \epsilon_i, \quad \forall u_i \in U,$$

where $\epsilon_{i,\tau}$ denotes the privacy budget consumption of u_i at time slot τ .

4 POPULATION DIVISION-BASED APPROACHES

Many studies [32, 36, 38] have shown that allocating the entire privacy budget to user groups, rather than dividing it among all users or time slots, can yield higher utility. Motivated by this observation, we adopt the population division paradigm and formalize frequency estimation under PLDP as the Personalized Frequency Oracle (PFO) protocol. We then analyze the estimation errors caused by population division in PFO. Finally, we propose two population division-based approaches: PLDP Population Distribution (PLPD) and PLDP Population Absorption (PLPA), which achieve accurate estimation by dynamically recycling population while satisfying PLDP constraints.

4.1 Personalized Frequency Oracle

Similar to Frequency Oracle (FO) [39] in the LDP model, frequency estimation under PLDP can also be encapsulated into a standard protocol. Here, we refer to this as the Personalized Frequency Oracle (PFO).

PFO consists of three functions: the *Perturbation* function PFO.P, the *Aggregate* function PFO.A and the *Estimation* function PFO.E. The function PFO.P randomizes the raw data into obfuscated data using the PLDP algorithm. PFO.A calculates the aggregate statistic for different groups of the obfuscated data. PFO.E is responsible for estimating the distribution of raw data.

We adopt Generalized Personalized Random Response (GPRR) as the personalized frequency oracle due to its simplicity and unbiasedness. Each user perturbs his/her value using GPRR with personalized privacy budget ϵ_i :

$$\text{PFO.P} : \forall y_i \in \Omega, \Pr[\mathcal{M}(x_i) = y_i] = \begin{cases} p_i = \frac{e^{\epsilon_i}}{e^{\epsilon_i} + d - 1}, & \text{if } y_i = x_i, \\ q_i = \frac{1}{e^{\epsilon_i} + d - 1}, & \text{otherwise.} \end{cases}$$

After receiving the obfuscated reports y_k from all users across different groups divided by distinct $\tilde{\epsilon}_k \in \tilde{\epsilon}$, the server computes the obfuscated statistic of each domain value $\omega_j \in \Omega$ for every group G_k :

$$\text{PFO.A} : \forall k \in [m], j \in [d], h_{k,j} = \frac{\text{Count}(y_{k,j})}{n \cdot g_k},$$

where n is the total number users and $g_k \in \mathbf{g}$ is the ratio of users requiring $\tilde{\epsilon}_k$. Thus, the obfuscated frequency list for each group G_k can be calculated as $\mathbf{h}_k = \langle h_{k,1}, h_{k,2}, \dots, h_{k,d} \rangle$. Subsequently, the server estimates the overall frequency distribution as:

$$\text{PFO.E: } \forall \hat{f}_j \in \hat{\mathbf{f}}, \hat{f}_j = \sum_{k=1}^m \alpha_k \cdot \frac{h_{k,j} - \tilde{q}_k}{\tilde{p}_k - \tilde{q}_k}, \quad (1)$$

where $\alpha_k = \frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\text{Var}[\hat{f}_{k',j}]}$ denotes the weight of group G_k , and

$\text{Var}[\hat{f}_{k,j}]$ represents the variance under ϵ_k -LDP [17] with

$$\text{Var}[\hat{f}_{k,j}] = \frac{\tilde{q}_k(1 - \tilde{q}_k)}{ng_k(\tilde{p}_k - \tilde{q}_k)^2} + \frac{1 - \tilde{p}_k - \tilde{q}_k}{ng_k(\tilde{p}_k - \tilde{q}_k)} \cdot f_j, \quad (2)$$

where $\hat{p}_k = \frac{e^{\tilde{\epsilon}_k}}{e^{\tilde{\epsilon}_k} + d - 1}$ and $\hat{q}_k = \frac{1}{e^{\tilde{\epsilon}_k} + d - 1}$.

Equation (2) shows that the variance depends on the true frequency f_j . However, since f_j is unknown, this definition is impractical for direct computation. To address this issue, we replace $\text{Var}[\cdot]$ with the average variance $\overline{\text{Var}}[\cdot]$, and redefine the weight

as $\alpha_k = \frac{1/\overline{\text{Var}}[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\overline{\text{Var}}[\hat{f}_{k',j}]}$, where

$$\overline{\text{Var}}[\hat{f}_{k,j}] = \frac{\tilde{q}_k(1 - \tilde{q}_k)}{ng_k(\tilde{p}_k - \tilde{q}_k)^2} + \frac{1 - \tilde{p}_k - \tilde{q}_k}{ng_k(\tilde{p}_k - \tilde{q}_k)} \cdot \frac{1}{d}.$$

Based on the modified weights, we can calculate the total variance in PLDP as:

$$\text{Var}[\hat{\mathbf{f}}] = 1 / \sum_{k=1}^m n / (d\lambda_k + \mu_k), \quad (3)$$

where $\lambda_k = \frac{\tilde{q}_k(1 - \tilde{q}_k)}{g_k(\tilde{p}_k - \tilde{q}_k)^2}$ and $\mu_k = \frac{1 - \tilde{p}_k - \tilde{q}_k}{g_k(\tilde{p}_k - \tilde{q}_k)}$. For more details of the modified weights and the calculation process for the total variance, please refer to Appendix 8.2 of our report [11]. We abbreviate the total variance in Equation (3) as $V_{\text{PLDP}}(\tilde{\epsilon}, \mathbf{g}, n, d)$.

4.2 Private Strategy Determination

The core idea of population division methods is to adaptively adjust the participating population at each time slot. Specifically, the method compares the privacy dissimilarity dis with the reporting error err . If $dis > err$, a new estimation is made using a new portion of the population; otherwise, the previous estimation is reused. Next, we define dis and err in detail.

Dissimilarity. The definition of dis in ϵ -PLDP is similar to that in the standard LDP [32], which is defined as

$$dis = \frac{1}{d} \sum_{j=1}^d (\hat{f}_{s,t}[j] - \hat{f}_l[j])^2 - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]].$$

Further details about the computation and analysis of dis are provided in Appendix 8.3 of our report [11]. In contrast, the definition of err in PLDP is more complex and differs substantially from that in ϵ -PLDP. Therefore, we need to redefine these variables accordingly.

Reporting Errors. Similar to population division methods under LDP [38], population division under PLDP can also be decomposed into two components: the *variance due to PLDP perturbation* and the *variance due to sampling*. The variance for PLDP perturbation, denoted as $V_{\text{PLDP}}(\tilde{\epsilon}, \mathbf{g}, n, d)$, has been shown in Equation (3).

Variance for Sampling. Given a sample of users U with size z , let X_j denote the number of occurrences of $\omega_j \in \Omega$ within this sample. Then X_j follows a Hypergeometric Distribution [20], and its variance can be expressed as

$$\text{Var}[X_j] = z \cdot f_j \cdot (1 - f_j) \cdot \frac{n - z}{n - 1}.$$

Accordingly, the variance of estimated frequency for Ω_j is:

$$\text{Var}[\hat{f}_j] = \frac{1}{z^2} \text{Var}[X_j] = \frac{f_j \cdot (1 - f_j)}{z} \cdot \frac{n - z}{n - 1}.$$

Thus, the variance of the entire frequency estimation is given by:

$$\text{Var}[\hat{\mathbf{f}}] = \sum_{j=1}^d \frac{f_j \cdot (1 - f_j)}{z} \cdot \frac{n - z}{n - 1} = \frac{n - z}{z(n - 1)} \left(1 - \sum_{j=1}^d f_j^2 \right). \quad (4)$$

When the domain size d is large, $\sum_{j=1}^d f_j^2 \approx 0$, and the upper bound of Equation (4) becomes:

$$\text{Var}^*[\hat{\mathbf{f}}] = \frac{n - z}{z(n - 1)}. \quad (5)$$

We abbreviate Equation (5) as $V_{\text{simpl}}(n, z)$.

Reporting Errors for population division under PLDP. The total variance combining both sampling and perturbation effects is expressed as:

$$\begin{aligned} V(\tilde{\epsilon}, \mathbf{g}, n, z, d) &= V_{\text{simpl}}(n, z) + V_{\text{PLDP}}(\tilde{\epsilon}, \mathbf{g}, z, d) \\ &= \frac{n - z}{z(n - 1)} + 1 / \sum_{k=1}^m z / (d\lambda_k + \mu_k), \end{aligned} \quad (6)$$

We defined the reporting error as the average variance across all values in the output domain, given by:

$$\begin{aligned} err(\tilde{\epsilon}, \mathbf{g}, n, z, d) &= \frac{1}{d} (V_{\text{simpl}}(n, z) + V_{\text{PLDP}}(\tilde{\epsilon}, \mathbf{g}, z, d)) \\ &= \frac{n - z}{dz(n - 1)} + 1 / \sum_{k=1}^m dz / (d\lambda_k + \mu_k), \end{aligned} \quad (7)$$

From Equation (7), we observe that, for fixed privacy requirements, a larger sampling size results in a smaller reporting error, which aligns with the intuitive expectation that using more population improves estimation accuracy. This observation provides the theoretical basis for designing adaptive mechanisms that dynamically allocate population to balance privacy and utility.

Building on this insight, we now introduce two population division-based methods, PLDP Population Distribution (PLPD) and PLDP Population Absorption (PLPA), which follow the main idea described above. These methods are intentionally simple extensions of non-personalized methods, designed to address the foundational challenges in the personalized setting. While they do not introduce entirely new techniques, they provide the necessary groundwork for more sophisticated methods in subsequent sections, offering a starting point for developing more complex personalized privacy solutions.

Both methods share the same private dissimilarity calculation module ($\mathcal{M}_{s,t}$), which consumes a fixed-size population $U_{s,t}$ at each time slot. However, they differ in the private publication module ($\mathcal{M}_{r,t}$), where each method adopts a distinct dynamic allocation

strategy for the publication population $U_{r,t}$ across the steam. We refer to $U_{s,t}$ as the *dissimilarity population*, and $U_{r,t}$ as the *publication population*.

4.3 PLDP Population Basic Solutions

Both PLPD and PLPA are built upon LDP Budget Distribution (LBD) and LDP Budget Absorption (LBA), respectively, as described in Reference [32]. At the initialization stage, both methods adopt a uniform population sampling size defined as $z = \min_{i \in [n]} \left\lfloor \frac{n}{w_i} \right\rfloor = \left\lfloor \frac{n}{w_{\max}} \right\rfloor$, and compute the dissimilarity dis and error err following the same principles as in PLPD and PLPA, while applying PFO for each estimation. Due to space limitations, detailed descriptions are deterred to Appendix 8.1 of our report [11].

4.4 Analysis

Time Complexity Analysis. Let d denote the value domain size, m the number of budget groups, and n the total number of users, satisfying $m \leq n$. Let $z_{\min}$ be the chosen sampling size at each time slot, with $z_{\min} \leq n/2$. We analyze the time complexity of PLPD and PLPA as follows.

Theorem 4.1. *The time complexity of both PLPD and PLPA is $O(n + d \cdot m)$.*

PROOF. Please refer to the detailed proof of Theorem 4.1 in Appendix 8.5.1 of our report [11]. $\square$

Privacy Analysis. When the window size is fixed (i.e., $w = w_{\max}$), each user in both PLPD and PLPA participates at most once within this window. Furthermore, every estimation is released through a PFO mechanism that satisfies ϵ -PLDP. Therefore, the following theorem holds for the privacy guarantee.

Theorem 4.2. *PLPD and PLPA satisfy (w, ϵ) -EPLDP.*

PROOF. Please refer to the detailed proof of Theorem 4.2 in Appendix 8.6.1 of our report [11]. $\square$

Utility Analysis. For simplicity, given the maximum window size $w_{\max}$, we assume that at most $s < w_{\max}$ new publications occur at time slots $\rho_1, \rho_2, \dots, \rho_s$, without any budget absorption from past time slots preceding the current window. In addition, there exists an equal number of skipped / nullified publications corresponding to these new publications. Let $\tilde{n}$ denote the user count list, where $\tilde{n}(k)$ represents the number of users requiring privacy budget list $\tilde{\epsilon}(k)$. Based on these settings, we provide the following two theorems for PLPD and PLPA, respectively.

Theorem 4.3. *The error upper bound of per time slot in PLPD is $\frac{1}{d^3} \left(\frac{2w_{\max}}{n-1} - \frac{1}{n-1} + 2w_{\max}A \right)^2 + \frac{4(2^s-1)}{s} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}$, where $A = \frac{d-1}{\min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}$.*

PROOF. The detailed proof of Theorem 4.3 is provided in Appendix 8.7.1 of our report [11]. $\square$

Theorem 4.4. *The error upper bound of per time slot in PLPA is $\frac{1}{d^3} \left(\frac{2w_{\max}}{n-1} - \frac{1}{n-1} + 2w_{\max}A \right)^2 + BC \cdot err_{nlf} - \frac{(B+1)C}{n-1} + \left(\frac{s}{n-1} + s \cdot A \right)$.*

Algorithm 1: Optimal Population Selection (OPS)

Input: The sampling size requirement list $z = \langle z_1, \dots, z_n \rangle$; the privacy budget requirement list $\epsilon = \langle \epsilon_1, \dots, \epsilon_n \rangle$; the value domain size d

Output: The optimal sampling size z_{opt} , the optimal error err_{opt} , the trasformed privacy budget vector ϵ'_{opt}

```

1 Initialize  $err_{opt}$  as the upper bound of error value;
2 Set  $\tilde{z} \leftarrow \text{Unique}(z)$  as the unique list of  $z$ ;
3 for  $\tilde{z} \in \tilde{z}$  do
4   Initialize  $\epsilon' \leftarrow \emptyset$ ;
5   for  $z_i \in z$  do
6     if  $z_i < \tilde{z}$  then
7       add  $\epsilon_i / \lceil \tilde{z} / z_i \rceil$  to  $\epsilon'$ ;
8     else
9       add  $\epsilon_i$  to  $\epsilon'$ ;
10  Set  $\tilde{\epsilon}' \leftarrow \text{Unique}(\epsilon')$ ;
11  Count the privacy requirement frequency  $g'$  of  $\tilde{\epsilon}'$ ;
12  Calculate  $err_{tmp} \leftarrow V(\tilde{\epsilon}', g', n, \tilde{z}, d)$  according to Equation (6);
13  if  $err_{tmp} < err_{opt}$  then
14     $err_{opt} \leftarrow err_{tmp}$ ;
15     $z_{opt} \leftarrow \tilde{z}$ ;
16     $\epsilon'_{opt} \leftarrow \epsilon'$ ;
17 return  $z_{opt}, err_{opt}, \epsilon'_{opt}$ ;

```

$\left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right)$, where $A = \frac{d-1}{\min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}$, $B = \frac{w_{\max} - s}{2s}$ and $C = \frac{s}{w_{\max}}$.

PROOF. The detailed proof of Theorem 4.4 is provided in Appendix 8.7.1 of our report [11]. $\square$

5 ENHANCED APPROACHES

PLPD and PLPA address the population assignment by unifying the window size to the maximum value, i.e., $w_{\max} = \max(w_1, w_2, \dots, w_n)$. However, these two methods face two main issues: (1) the use of maximum window size results in a minimum sampling size at each time slot, leading to large sampling errors; (2) in personalized local different privacy settings, data with higher privacy protection (i.e., smaller ϵ) can reduce the accuracy of the overall estimation.

To address the first issue, we propose the Optimal Population Selection (OPS) method, which selects an optimal sampling size to minimize the reporting error while maintaining privacy levels through budget division. For the second issue, we propose a technique to re-disturb high-utility data and combine it with low-utility data to enhance the accuracy of the latter.

5.1 Optimal Population Selection

In personalized local differential privacy settings, the optimal sampling size strategy should take into account both LDP error and sampling error. Thus, we need to determine the optimal sampling size z_{opt} that minimizes the total error. This process is illustrated in Algorithm 1.

OPS first calculates the unique sampling sizes $\tilde{z}$. Then it iterates over all sampling size in $\tilde{z}$ to find the optimal one that minimizes the reporting error (Lines 3–16). To maintain privacy protection, during the iteration, the privacy budget ϵ_i of users whose sampling size

z_i is smaller than the current $\tilde{z}$ are divided into $\lceil \tilde{z}/z_i \rceil$ shares. The privacy budget is then updated as a share of size $\epsilon_i / \lceil \tilde{z}/z_i \rceil$ (Lines 5–9). OPS calculates the statistic g' of distinct new privacy budgets $\tilde{\epsilon}'$ and records the minimum error err_{opt} with the corresponding optimal sampling size z_{opt} (Lines 10–15).

An example of OPS is provided in Example 2.

Example 2. Assume there are 10 users with sampling sizes $z = \langle 3, 5, 9, 8, 9, 5, 6, 5, 8, 9 \rangle$ and personalized privacy budgets $\epsilon = \langle 0.1, 0.4, 0.4, 0.1, 0.4, 0.4, 0.8, 0.8, 0.8, 0.4 \rangle$ with $d = 2$. Then, the distinct sampling sizes are $\tilde{z} = \langle 3, 5, 6, 8, 9 \rangle$. OPS iterates over $\tilde{z} \in \tilde{z}$, dividing the privacy budgets with smaller sampling sizes and calculating the corresponding error. For $\tilde{z} = 6$, users with $z = 3$ and $z = 5$ divide their budgets, resulting in $\epsilon' = \langle 0.05, 0.2, 0.4, 0.1, 0.4, 0.2, 0.8, 0.4, 0.8, 0.4 \rangle$ and $err = 0.988$. The errors for all $\tilde{z}$ are $\langle 1.121, 0.596, 0.988, 0.721, 1.124 \rangle$, yielding the optimal sampling size $z_{opt} = 5$ with minimum error $err = 0.596$.

5.2 Utility Improvement

To improve utility in private calculations (e.g., dis or new estimations), we propose using obfuscated data with lower privacy levels (higher utility) to enhance the accuracy of data with higher privacy levels (lower utility). This idea has been explored in privacy protection studies [17, 26] and learning-based frameworks [33, 34]. We extend the re-perturbation method from [17] to GPRR settings and propose the GPRR re-perturbation mechanism.

Theorem 5.1. For any two budget group data $BGD_L = (\epsilon_L, U_L, Y_L)$ and $BGD_H = (\epsilon_H, U_H, Y_H)$ under GPRR with $\epsilon_L < \epsilon_H$, the variance of BGD_L can be reduced by

$$\frac{z_H}{z_L(z_L + z_H)} \left(\frac{dq_L(1 - q_L)}{(p_L - q_L)^2} + \frac{1 - p_L - q_L}{p_L - q_L} \right)$$

through updating BGD_L to $BGD'_L = (\epsilon_L, U_L \cup U_H, Y_L \cup Y'_H)$, where Y'_H is the re-perturbed output of Y_H with probability:

$$\forall y' \in Y', \Pr[\text{RP}(y) = y'] = \begin{cases} \beta, & \text{if } y' = y, \\ \gamma, & \text{otherwise,} \end{cases} \quad (8)$$

where $\beta = \frac{dp_L - p_L + p_H - 1}{dp_H - 1}$ and $\gamma = \frac{p_H - p_L}{dp_H - 1}$.

PROOF. Please refer to the detailed proof of Theorem 5.1 in Appendix 8.4 of our report [11]. $\square$

Based on the re-perturbation method, we propose an enhanced utility approach called Personalized Utility Enhancement (PUE) shown in Algorithm 2. In the PUE process, privacy budgets, populations and obfuscated counts are sorted by the privacy budgets. PUE iterates over all the privacy budget ϵ_j and applies re-perturbation to the obfuscated counts whose privacy budgets are larger than ϵ_i (Line 6).

Example 3. Assume the value domain is $\omega = \{0, 1\}$ and there are 5 groups $G_1, G_2, \dots, G_5$ with sorted privacy budgets $\tilde{\epsilon} = \langle 0.2, 0.3, 0.5, 0.6, 0.8 \rangle$. Assume the obfuscated statistics of groups G_3, G_4 and G_5 are $\{40, 60\}, \{70, 30\}$ and $\{60, 50\}$, where each group adopts GRR as its base mechanism. Assume the current enhanced group is G_3 ($\tilde{\epsilon}_3 = 0.6$). Then, PUE re-perturbs all users in G_4 and G_5 . For $u_1 \in G_4$ with obfuscated data $y_1 = 1$, the re-perturbed data is 1 with

Algorithm 2: Personalized Utility Enhancement (PUE)

Input: A privacy budget list $\tilde{\epsilon} = \langle \tilde{\epsilon}_1, \tilde{\epsilon}_2, \dots, \tilde{\epsilon}_m \rangle$; a perturbed data list $Y = \langle Y_1, Y_2, \dots, Y_m \rangle$

Output: The enhanced re-perturbed data $\hat{Y}$

- 1 Construct ordered budget group data
EBGD = $\langle BGD_1, BGD_2, \dots, BGD_m \rangle$ sorted by $\tilde{\epsilon}$ in ascending order;
 - 2 Initialize $\hat{Y} \leftarrow \emptyset$;
 - 3 **for** $i \in [m]$ **do**
 - 4 Initialize $\hat{Y}_i \leftarrow Y_i$;
 - 5 **for** $j \in [i + 1, m]$ **do**
 - 6 Get re-perturbed value $Y'_j \leftarrow \text{RP}(Y_j, \tilde{\epsilon}_j, \tilde{\epsilon}_i)$ by Equation 8;
 - 7 $\hat{Y}_i \leftarrow \hat{Y}_i \cup Y'_j$;
 - 8 $\hat{Y} \leftarrow \hat{Y} \cup \hat{Y}_i$
 - 9 **return** $\hat{Y}$;
-

probability $\beta_{4,3} = \frac{dp_3 - p_3 + p_4 - 1}{dp_4 - 1} = 0.65$ and 0 with probability $\gamma_{4,3} = \frac{p_4 - p_3}{dp_4 - 1} = 0.35$. For $u_2 \in G_5$ with $y_2 = 0$, the re-perturbed data is 1 with probability $\gamma_{5,3} = \frac{p_5 - p_3 + p_4 - 1}{dp_5 - 1} = 0.18$ and 0 with probability $\beta_{5,3} = 0.82$.

5.3 PLDP Population Enhanced Solutions

Building on the OPS and PUE methods, we propose two enhanced solutions: PLDP Population Distribution Plus (PLPD⁺) and PLDP Population Absorption Plus (PLPA⁺). These methods calculate the optimal sampling population while ensuring privacy via OPS, and enhance utility by improving personalized responses using PUE. After frequency estimation, we apply Simplex Projection [8] to ensure the estimated frequencies are valid probability distributions. **PLDP Population Distribution Plus.** Algorithm 3 shows the process of PLPD⁺. In PLPD⁺, the population is treated as a resource and divided among the slots in each window. It is further split into two subsets, $U_{s,t}$ and $U_{r,t}$, for calculations in two sub-mechanisms: $\mathcal{M}_{s,t}$ and $\mathcal{M}_{r,t}$. $\mathcal{M}_{s,t}$ consumes $U_{s,t}$ to calculate dissimilarity dis , which is compared with the current error err to decide whether a new obfuscated statistic should be published. $\mathcal{M}_{s,t}$ uses $U_{r,t}$ to publish a new obfuscated data when $dis > err$.

For the process of $\mathcal{M}_{s,t}$, PLPD⁺ first calculates the population sampling size list z for all users (Line 2). Using these personalized sampling sizes, PLPD⁺ applies OPS to obtain the optimal size z_{opt} while ensuring privacy protection through budget division (Line 3). Next, it samples $U_{s,t}$ from the available user set U_A (Line 4) and applies the PFO protocol to calculate the dissimilarity (Line 6–10). Unlike PLPD, PLPD⁺ applies PUE to improve the utility of disturbed data from PFO (Line 7).

For the process of $\mathcal{M}_{r,t}$, PLPD⁺ calculates the remaining population resource $n_{r,opt}$ for new publication within the window w_{opt} (Line 12). It reserves half of $n_{r,opt}$ (i.e., $n_{pp,opt}$) for the new publication and leaves the other half for future use (Line 13). Based on the reserved population, it calculates the error err for the new publication (Line 16). Unlike traditional LDP setting [32], here, PLPD⁺ samples the population for the new publication before the dissimilarity-error comparison to improve the accuracy of the error calculation without introducing additional privacy leakage (Line 14). Then, it calculates the current error err according to Equation (7) (Line 15–16) and

Algorithm 3: PLDP Population Distribution Plus

Input: Available user set U_A , privacy requirement (w, ϵ) of all Users U , data domain size d , historical data publication $(\hat{f}_1, \hat{f}_2, \dots, \hat{f}_{t-1})$

Output: $\hat{f}_t$

- 1 Calculate the distinct privacy budget $\tilde{\epsilon}$ with the statistic g ;
// sub-mechanism $M_{s,t}$
- 2 Calculate the sampling size $z \leftarrow \left\lceil \left\lfloor \frac{n}{2w_1} \right\rfloor, \left\lfloor \frac{n}{2w_2} \right\rfloor, \dots, \left\lfloor \frac{n}{2w_n} \right\rfloor \right\rceil$;
- 3 Get optimal population size and error
 $z_{opt}, err_{opt}, \epsilon_{opt} \leftarrow OPS(z, \epsilon, d)$ by Algorithm 1;
- 4 Sample user set $U_{s,t}$, whose data is $D_{s,t}$ and privacy budget list is $\epsilon'_{opt} \subseteq \epsilon_{opt}$, from U_A with the size of z_{opt} and remove $U_{s,t}$ from U_A , i.e., $U_A = U_A \setminus U_{s,t}$;
- 5 $\tilde{\epsilon}'_{opt} \leftarrow \text{Unique}(\epsilon'_{opt})$;
- 6 Get report $Y_{s,t} \leftarrow \text{PFO.P}(D_{s,t})$ with privacy budget list ϵ'_{opt} from users in $U_{s,t}$;
- 7 Calculate the enhanced report $\hat{Y}_{s,t} \leftarrow \text{PUE}(\tilde{\epsilon}'_{opt}, Y_{s,t})$ by Algorithm 2;
- 8 Get $h_{s,t,k} \leftarrow \text{PFO.A}(\hat{Y}_{s,t})$ for each group G_k ;
- 9 Estimate $\hat{f}_{s,t} \leftarrow \text{PFO.E}(h_{s,t}, \tilde{\epsilon}'_{opt})$;
- 10 Calculate $dis \leftarrow \frac{1}{d} \sum_{j=1}^d (\hat{f}_{s,t}[j] - \hat{f}_t[j])^2 - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]]$;
// sub-mechanism $M_{r,t}$
- 11 Calculate the optimal window size $w_{opt} \leftarrow n/(2z_{opt})$;
- 12 Calculate the optimal remaining population size
 $n_{r,opt} \leftarrow n/2 - \sum_{\tau=t-w_{opt}+1}^{t-1} |U_{r,\tau}|$;
- 13 Set the optimal number of potential publication users
 $n_{pp,opt} \leftarrow \lfloor n_{r,opt}/2 \rfloor$;
- 14 Sample a user set $U_{r,t}$ from U_A with the size of $|U_{r,t}| = n_{pp,opt}$ and privacy budget list $\epsilon''_{opt} \subseteq \epsilon_{opt}$;
- 15 Calculate the budget group user statistic $g_{pp,opt}$ of
 $\tilde{\epsilon}''_{opt} = \text{Unique}(\epsilon''_{opt})$ by $U_{r,t}$;
- 16 Calculate the potential reporting error
 $err \leftarrow err(\tilde{\epsilon}''_{opt}, g_{pp,opt}, n, n_{pp,opt}, d)$ by Equation (7);
- 17 **if** $dis > err$ and $n_{pp,opt} \geq \text{then}$
- 18 Remove $U_{r,t}$ from U_A , i.e., $U_A \leftarrow U_A \setminus U_{r,t}$;
- 19 Get report $Y_{r,t} \leftarrow \text{PFO.P}(D_{r,t})$ with privacy budget list ϵ''_{opt} from users in $U_{r,t}$;
- 20 Calculate the enhanced report $\hat{Y}_{r,t} \leftarrow \text{PUE}(\tilde{\epsilon}''_{opt}, Y_{r,t})$;
- 21 Get $h_{r,t,k} \leftarrow \text{PFO.A}(\hat{Y}_{r,t})$ for each group G_k ;
- 22 Calculate $\hat{f}_{r,t} \leftarrow \text{PFO.E}(h_{r,t}, \tilde{\epsilon}''_{opt})$;
- 23 **else**
- 24 Set $\hat{f}_{r,t} \leftarrow \hat{f}_{r,t-1}$;
- 25 **if** $t \geq w_{opt}$ **then**
- 26 $U_A \leftarrow U_A \cup U_{s,t-w_{opt}+1} \cup U_{r,t-w_{opt}+1}$;
- 27 **return** $\hat{f}_{r,t}$.

compares it with the dissimilarity dis (Line 17–24). If $dis > err$, it indicates that the difference between the current and previous estimation is large enough to warrant a new publication. In this case, PLPD⁺ removes the sampled $U_{r,t}$ from the candidate population (Line 18) and applies enhanced PFO to generate a new estimation $\hat{f}_{r,t}$ (Line 19–22). If $dis \leq err$, PLPD⁺ discards the sampled $U_{r,t}$ and approximates the current estimation as the one at the previous time slot (Line 24). Finally, PLPD⁺ recycles the population at the $w_{opt} - 1$ time slot (Line 26).

We give an example for the procedure of PLPD⁺ in Example 4.

Example 4. Consider 5 locations $\{A, B, C, D, E\}$ and $n = 2000$ users $u_1, u_2, \dots, u_{2000}$. Each user selects a privacy budget from $\{0.2, 0.4, 0.6, 0.8\}$ and window size from $\{1, 2, 3, 4\}$. Thus, the distinct privacy budget list is $\tilde{\epsilon} = \langle 0.2, 0.4, 0.6, 0.8 \rangle$, with corresponding probability vectors $\tilde{q} = \langle 0.19, 0.18, 0.17, 0.16 \rangle$ and $\tilde{p} = \langle 0.23, 0.27, 0.31, 0.36 \rangle$. Table 4(a) reports the count of each (w, ϵ) pair, and the resulting budget-group frequency vector is $g = \langle 0.1305, 0.3645, 0.244, 0.261 \rangle$. The distinct sampling size list is $\tilde{z} = \langle 1000, 500, 333, 250 \rangle$. Running OPS yields error values $err = \langle 0.042, 0.035, 0.0329, 0.0349 \rangle$. Thus, $z_{opt} = 333$. Since $\tilde{w}_4 = 4$ corresponds to $z = 250 < 333$, budgets for users with $\tilde{w}_4 = 4$ are divided into $\lceil \tilde{z}_3/\tilde{z}_4 \rceil = \lceil 333/250 \rceil = 2$ shares, producing new budget values 0.1, 0.2, 0.3, 0.4 (shown in Table 4(b)).

At time slot $t = 1$, the available set is initialized as $U_A = U$. PLPD⁺ samples $z_{opt} = 333$ users, denoted as $U_{s,1} = \{u_1, u_2, \dots, u_{333}\}$, for dissimilarity estimation and updates $U_A = U_A \setminus U_{s,1} = \{u_{334}, u_{335}, \dots, u_{2000}\}$. Users in $U_{s,1}$ report via PFO.P. Assume the true statistic is $f_{s,1} = \langle 0.183, 0.237, 0.174, 0.228, 0.177 \rangle$, and the perturbed statistic is $\tilde{f}_{s,1} = \langle 0.204, 0.186, 0.189, 0.204, 0.216 \rangle$. After applying PUE and PFO.A, the effective user count increases to 1356 and the re-perturbed statistic becomes $\tilde{h}_{s,1} = \langle 0.204, 0.198, 0.190, 0.198, 0.209 \rangle$. PFO.E yields $\hat{f}_{s,1} = \langle 0.301, 0.194, -0.047, 0.336, 0.215 \rangle$, which is normalized by Simplex Projection to obtain $\langle 0.29, 0.182, 0, 0.324, 0.204 \rangle$. The weight vector for distinct privacy budget list $\tilde{\epsilon} = \langle 0.1, 0.2, 0.3, 0.4, 0.6, 0.8 \rangle$ is $\alpha = \langle 0.015, 0.062, 0.117, 0.207, 0.286, 0.313 \rangle$, and $\sum_{j=1}^d \text{Var}[\hat{f}_{s,1}[j]] = 0.085$, giving $dis = 0.041$. The potential publication size is $n_{pp,opt} = \lfloor n/2 \rfloor = 500$, and the reporting error as $err = 0.021$. Since $dis > err$, the algorithm samples 500 users from U_A for a new publication. Assume the sampling set is $U_{r,1} = \{u_{334}, u_{335}, \dots, u_{833}\}$. Assume $f_{r,1} = \langle 0.2, 0.206, 0.196, 0.182, 0.216 \rangle$, $\tilde{f}_{r,1} = \langle 0.206, 0.19, 0.194, 0.23, 0.18 \rangle$, and $h_{r,1} = \langle 0.23, 0.19, 0.19, 0.21, 0.18 \rangle$. After re-perturbing, $\tilde{h}_{r,1} = \langle 0.209, 0.196, 0.208, 0.201, 0.187 \rangle$, and the final estimation is $\hat{f}_{r,1} = \langle 0.21, 0.108, 0.322, 0.281, 0.079 \rangle$ with $\alpha = \langle 0.016, 0.064, 0.12, 0.21, 0.28, 0.313 \rangle$. As $t < w_{max}$, no recycling occurs yet.

At $t = 2$, PLPD⁺ again samples 333 users for dis calculation. Assume these 333 users are $U_{s,2} = \{u_{834}, u_{835}, \dots, u_{1166}\}$ and $dis = 0.041$. Then the remaining population for publication is $n_{r,opt} = 2000/2 - 500 = 500$, hence $n_{pp,opt} = \lfloor n_{r,opt}/2 \rfloor = 250$. The method samples $n_{pp,opt} = 125$ candidate users for a new publication and calculates the reporting error as $err = 0.05$. Since $dis \leq err$, the publication is aborted and $\hat{f}_{r,2} = \hat{f}_{r,1}$.

Assume a new publication happens at $t = 3$. Then the sampling population consumption is still 333, and the new publication population becomes $\lfloor (n/2 - 500 - 0)/2 \rfloor = 250$, leaving $|U_A| = 251$ after publication. Assume $U_A = \{u_{1170}, u_{1171}, \dots, u_{2000}\}$. Since $t = 3 \geq w_{opt}$, PLPD⁺ recycles the population consumed at $t = 1$, i.e., $U_A = U_A \cup U_{s,1} \cup U_{r,1} = \{u_1, u_2, \dots, u_{833}, u_{1170}, u_{1171}, \dots, u_{2000}\}$, resulting in $|U_A| = 1084$. The procedure continues similarly for subsequent time slots.

PLDP Population Absorption Plus. PLPD⁺ reserves half of the remaining population for future publications, making it suitable for slow, gradual changes in data streams. However, when the data stream encounters occasional but sudden, PLPD⁺ may fail to provide

Table 4: An example for the user counts of window size-privacy budget pairs.

(a) The original data counts					
$\epsilon \backslash w$	1	2	3	4	
0.2	29	59	92	56	
0.4	98	176	273	207	
0.6	62	124	188	114	
0.8	64	115	202	141	

(b) The optimal new data counts					
$\epsilon \backslash w$	1	2	3	4	
0.1	0	0	0	56	
0.2	29	59	92	207	
0.3	0	0	0	114	
0.4	98	176	273	141	
0.6	62	124	188	0	
0.8	64	115	202	0	

accurate estimation. To address this limitation, we propose PLDP Population Absorption Plus (PLPA⁺) as follows.

Algorithm 4: PLDP Population Absorption Plus

Input: Available user set U_A , privacy requirement (w, ϵ) of all Users U , data domain size d , historical data publication $(\hat{f}_1, \hat{f}_2, \dots, \hat{f}_{t-1})$

Output: $\hat{f}_t$

- 1 Calculate the distinct privacy budget $\tilde{\epsilon}$ with the statistic g ;
// sub-mechanism $\mathcal{M}_{s,t}$
- 2 The same as Line 2-10 in Algorithm 3;
// sub-mechanism $\mathcal{M}_{r,t}$
- 3 Calculate the optimal window size $w_{opt} \leftarrow n/(2z_{opt})$;
- 4 Calculate the time slots to be nullified $t_N = \lfloor \frac{|U_{r,l}|}{z_{opt}} \rfloor - 1$;
- 5 **if** $t - l \leq t_N$ **then**
- 6 $\hat{f}_t \leftarrow \hat{f}_{t-1}$;
- 7 **else**
- 8 Calculate time slots that can be absorbed $t_A = t - l - t_N$;
- 9 Set the number of potential publication users
 $n_{pp,opt} \leftarrow z_{opt} \cdot \min(t_A, w_{opt})$;
- 10 Sample a user set $U_{r,t}$ from U_A with the size of $|U_{r,t}| = n_{pp,opt}$
 and privacy budget list $\epsilon''_{opt} \subseteq \epsilon_{opt}$
- 11 Calculate the budget group user statistic $g_{pp,opt}$ of
 $\tilde{\epsilon}''_{opt} = \text{Unique}(\epsilon''_{opt})$ by $U_{r,t}$.
- 12 Calculate the potential reporting error
 $err \leftarrow err(\tilde{\epsilon}''_{opt}, g_{pp,opt}, n, n_{pp,opt}, d)$ by Equation (7);
- 13 **if** $dis > err$ **then**
- 14 Remove $U_{r,t}$ from U_A , i.e., $U_A \leftarrow U_A \setminus U_{r,t}$;
- 15 Get report $Y_{r,t} \leftarrow \text{PFO.P}(D_t)$ with privacy budget list ϵ''_{opt}
 from users in $U_{r,t}$;
- 16 Calculate the enhanced report $\hat{Y}_{r,t} \leftarrow \text{PUE}(\tilde{\epsilon}''_{opt}, Y_{r,t})$;
- 17 Get $h_{r,t,k} \leftarrow \text{PFO.A}(\hat{Y}_{r,t})$ for each group G_k ;
- 18 Calculate $\hat{f}_{r,t} \leftarrow \text{PFO.E}(h_{r,t}, \tilde{\epsilon}''_{opt})$;
- 19 **else**
- 20 Set $\hat{f}_{r,t} \leftarrow \hat{f}_{r,t-1}$;
- 21 **if** $t \geq w_{opt}$ **then**
- 22 $U_A \leftarrow U_A \cup U_{s,t-w_{opt}+1} \cup U_{r,t-w_{opt}+1}$;
- 23 **return** $\hat{f}_{r,t}$.

The main idea of PLPA⁺ is to pre-assign an equal population share for each time slot and absorb the unused population from previous time slots, meanwhile nullifying the population for future time slots. This approach enhances the utility of the current publication. The process of PLPA⁺ is shown in Algorithm 4.

PLPA⁺ also consists of two sub-mechanisms: $\mathcal{M}_{s,t}$ and $\mathcal{M}_{r,t}$, where $\mathcal{M}_{s,t}$ calculates the dissimilarity dis and $\mathcal{M}_{r,t}$ publishes new

estimations. The mechanism $\mathcal{M}_{s,t}$ in PLPA⁺ is the same as in Algorithm 3.

For $\mathcal{M}_{r,t}$, PLPA⁺ calculates the time length t_N nullified by the last time slot t_l (Line 4), and checks whether the current time slot is within this period t_N . If it is, the current publication is approximated by the previous one (Line 6). Otherwise, PLPA⁺ compares dis and err to decide whether to publish a new obfuscated estimation (Line 8–20).

Specifically, PLPA⁺ calculates the time slot length t_A out of the nullified period t_N but not after the current time slot, determining the maximum length that can be absorbed by the current time slot. Then, PLPA⁺ calculates the population size $n_{pp,opt}$ that can be absorbed (Line 9), samples $n_{pp,opt}$ users $U_{r,t}$, and calculates the corresponding error err (Line 10–11). If $dis > err$, PLPA⁺ updates the population candidate set U_A (Line 14) and applies PFO enhanced by PUE to publish a new estimation (Line 15–18). Otherwise, it approximates the current publication as the previous one (Line 20). Finally, PLPA⁺ recycles the population outside the window size w_{opt} (Line 22).

The process of PLPA⁺ is illustrated in Example 5.

Example 5. Consider the data in Example 4. At time slot 1, assume PLPA⁺ consumes population $\{u_1, u_2 \dots u_{333}\}$ to calculate the dissimilarity dis . It calculates the nullified time length $t_N = -1$. Since $t - l = 1 > t_N$, PLPA⁺ compares dis with err to decide whether to publish a new estimation. Specifically, it calculates $t_A = 2$ and sets $n_{pp,opt} = 333 \times 2 = 666$. After sampling 666 users $U_{r,1}$, it calculates the error err . Assume $U_{r,1} = \{u_{334}, u_{335}, \dots, u_{999}\}$ and $dis > err$. Then PLPA⁺ publishes a new estimation $\hat{f}_{r,1}$ using enhanced PFO.

At time slot 2, dis calculation still consumes $z_{opt} = 333$ users $U_{s,2} = \{u_{1000}, u_{1001}, \dots, u_{1332}\}$. PLPA⁺ calculates the nullified time length $t_N = 666/333 - 1 = 1$. Since $t - l = 1 \leq t_N$, the current estimation $\hat{f}_{r,2}$ is approximated as $\hat{f}_{r,1}$.

At time slot 3, assume there is a new estimation publication. Then it consumes 333 users $U_{s,3} = \{u_{1333}, u_{1334}, \dots, u_{1665}\}$ for dis calculation. Here, $t_N = 666/333 - 1 = 1$ and $t_A = 3 - 1 - 1 = 1$. Thus, $n_{pp,opt}$ is set to 333. After sampling population $U_{r,3} = \{u_{1666}, u_{1667}, \dots, u_{1998}\}$, PLPA⁺ recycles $U_{s,1}$ and $U_{r,1}$, updating the candidate population set as $U_A = \{u_1, u_2, \dots, u_{999}, u_{1999}, u_{2000}\}$.

The process continues similarly for subsequent time slots.

5.4 Analysis

Time Complexity Analysis. Let d denote the value domain size, m the number of budget groups, and n the total number of users, satisfying $m \leq n$. Let z_{opt} be the optimal sampling size at each time slot, with $z_{opt} \leq n/2$. We analyze the time complexity of PLPD and PLPA as follows.

Theorem 5.2. The time complexity of both PLPD⁺ and PLPA⁺ is $O(n \cdot m + d \cdot m)$.

PROOF. Please refer to the detailed proof of Theorem 5.2 in Appendix 8.5.2 of our report [11]. $\square$

Privacy Analysis. When the optimal window size w_{opt} is determined, each user in both PLPD⁺ and PLPA⁺ appears at most once within any w_{opt} size window. For any user u_i with a personalized

window size requirement $w_i > w_{opt}$, the privacy budget ϵ_i is divided into $\lceil w_i/w_{opt} \rceil$ portions, ensuring the total privacy budget consumption within a w_i -length window does not exceed ϵ_i . In addition, each estimation is published through the PFO protocol under ϵ -PLDP. Therefore, the following theorem holds for the privacy guarantee of both methods.

Theorem 5.3. *PLPD⁺ and PLPA⁺ satisfy (w, ϵ) -EPLDP.*

PROOF. Please refer to the detailed proof of Theorem 5.3 in Appendix 8.6.2 of our report [11]. $\square$

Utility Analysis. For simplicity, let w_{opt} denote the optimal window size from the OPS process. We assume that at most $s_{opt} < w_{opt}$ new publications occur at time slots $\rho_1, \rho_2, \dots, \rho_{opt}$, without budget absorption from any past time slots outside the current window. Besides, we assume each new publication corresponds to the same count of skipped / nullified publications. Let $\tilde{\epsilon}_{opt}$ denote the new privacy budget list after applying OPS, and $\tilde{n}_{opt}$ represent the user count list, where $\tilde{n}_{opt}(k)$ indicates the number of users requiring privacy budgets $\tilde{\epsilon}_{opt}(k)$. Let $\hat{n}_{max}(\tilde{\epsilon}_{opt})$ denote the number of users requiring the maximum privacy budget in $\tilde{\epsilon}_{opt}$. Under these definitions, we present the following two theorems for PLPD⁺ and PLPA⁺, respectively.

Theorem 5.4. *The error upper bound of per time slot in PLPD⁺ is $\frac{1}{d^3} \left(\frac{2w_{opt}}{n-1} - \frac{1}{n-1} + 2w_{opt}A \right)^2 + \frac{4(2^{s_{opt}}-1)}{s_{opt}} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}$, where $A = \frac{d-1}{\hat{n}_{max}(\tilde{\epsilon}_{opt})} \cdot \frac{2e^{\min(\tilde{\epsilon}_{opt})} + d-2}{(e^{\min(\tilde{\epsilon}_{opt})} - 1)^2}$.*

PROOF. Please refer to the detailed proof of Theorem 5.4 in Appendix 8.7.2 of our report [11]. $\square$

Theorem 5.5. *The error upper bound of per time slot in PLPA⁺ is $\frac{1}{d^3} \left(\frac{2w_{opt}}{n-1} - \frac{1}{n-1} + 2w_{opt}A \right)^2 + BC \cdot err_{nlf} - \frac{(B+1)C}{n-1} + \left(\frac{s_{opt}}{n-1} + s_{opt} \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right)$, where $A = \frac{d-1}{\hat{n}_{max}(\tilde{\epsilon}_{opt})} \cdot \frac{2e^{\min(\tilde{\epsilon}_{opt})} + d-2}{(e^{\min(\tilde{\epsilon}_{opt})} - 1)^2}$, $B = \frac{w_{opt} - s_{opt}}{2s_{opt}}$ and $C = \frac{s_{opt}}{w_{opt}}$.*

PROOF. Please refer to the detailed proof of Theorem 5.5 in Appendix 8.7.2 of our report [11]. $\square$

6 EXPERIMENTS

We conduct experiments to evaluate our proposed mechanisms, comparing them with state-of-the-art non-personalized LDP methods for data streams [32]. Additionally, we perform an ablation study to assess the effectiveness of the key components OPS and PUE by removing each module and replacing it with its baseline mechanism.

6.1 Datasets

Real Datasets. We employ two real-world datasets, namely *Taxi* and *Foursquare*, to evaluate the performance of our PLPD⁺ and PLPA⁺.

Taxi. This dataset records real-time trajectories of 10,357 taxis in Beijing from February 2 to February 8, 2008. After cleaning, 14,859,377 valid records from $n = 10,269$ taxis are retained, and

Table 5: Experimental settings.

Parameters	Values
static privacy budget ϵ	0.5, 1.0, 1.5 , 2.0, 2.5
static window size w	40, 80, 120 , 160, 200
personalized privacy budget ϵ_i	$\epsilon, \dots, 2.0, 2.5$
personalized window size w_i	40, 80, $\dots, w$
population lower bound n_{min}	10

locations are mapped into $d = 5$ areas. We sample the trajectories every minute, resulting in $T = 8,889$ time slots.

Foursquare. This dataset consists of 33,278,683 Foursquare check-ins from 266,909 users collected between April 2012 and September 2013. We map all venues into $d = 5$ discrete categories and convert the data into $n = 13,216$ data streams with $T = 7,649$ time slots.

Synthetic Datasets. We generate three synthetic binary data streams based on different probabilistic sequence models to evaluate our mechanisms under controlled conditions. Each stream involves $n = 10,000$ users over $T = 10,000$ time slots. At each time step t , each user's binary value equals 1 with probability p_t (and 0 otherwise), where $(p_1, p_2, \dots, p_T)$ follows the generation rules defined below.

TLNS. In this dataset, the probability evolves dynamically as $p_t = p_{t-1} + \mathcal{N}(0, Q)$, where $\mathcal{N}(0, Q)$ denotes Gaussian noise with variance Q and standard deviation $\sqrt{Q} = 0.0025$. The initial probability is $p_0 = 0.05$, and all p_t values are clipped to the range $[0, 1]$ (i.e., set $p_t = 0$ if $p_t < 0$ and $p_t = 1$ if $p_t > 1$) to ensure validity.

Sin. In the Sin dataset, the probability follows a sinusoidal trend, defined as $p_t = A \sin(\omega t) + h$, where $A = 0.05$, $\omega = 0.01$, and $h = 0.075$.

Log. In the Log dataset, the probability follows a logistic growth pattern, defined as $p_t = \frac{A}{1 + e^{-bt}}$, where $A = 0.25$ and $b = 0.01$.

6.2 Experiment Setup

Compared Algorithms. We compare the enhanced personalized methods, PLPD⁺ and PLPA⁺, with their basic counterparts: PLPD and PLPA. Additionally, we also compare the above four personalized solutions with two non-personalized methods: LDP Budget Distribution (LBD) and LDP Budget Absorption (LBA) [32]. Each experiment is repeated ten times using different random seeds ranging from 0 to 9, and we report the average results.

Parameter Settings. In non-personalized settings (i.e., LBD and LBA), ϵ and w represent the privacy budget and window size, respectively. We vary ϵ from 0.5 to 1.5 and w from 40 to 200. Following the Reference [10], we set the lower bound of all users' personalized privacy budgets to ϵ and the upper bound of their personalized window sizes to w .

Table 5 summaries all experimental parameters, where default values are highlighted in bold. All experiments are conducted in Java on an Intel(R) Xeon(R) Silver 4210R CPU @ 2.4GHz with 128 RAM.

Performance Metrics. We evaluate each method in terms of running time and data utility. Data utility is measured as the *Average Mean Square Error (AMSE)*, defined as follows.

$$AMSE = \frac{1}{T} \sum_{\tau=1}^T MSE_{\tau} = \frac{1}{dT} \sum_{\tau=1}^T \sum_{j=1}^d (\hat{f}_{j,\tau} - f_{j,\tau})^2.$$

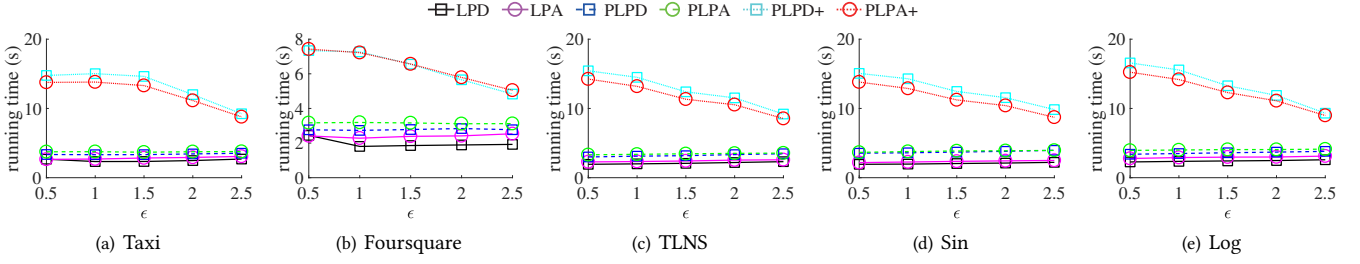


Figure 2: The running time with ϵ varied.

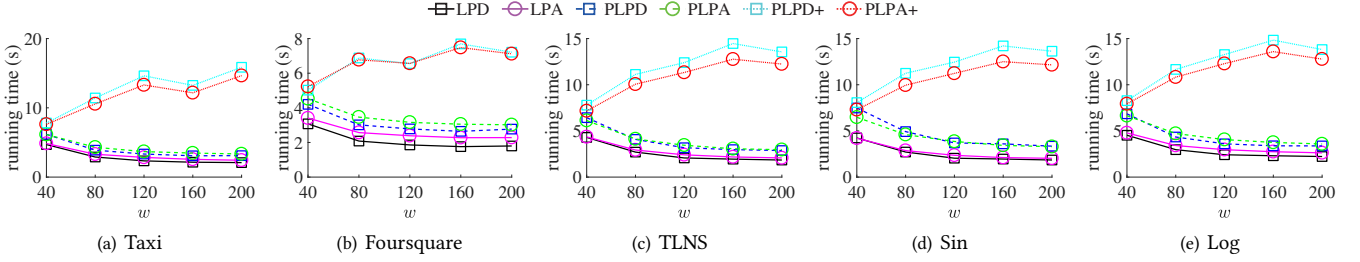


Figure 3: The running time with w varied.

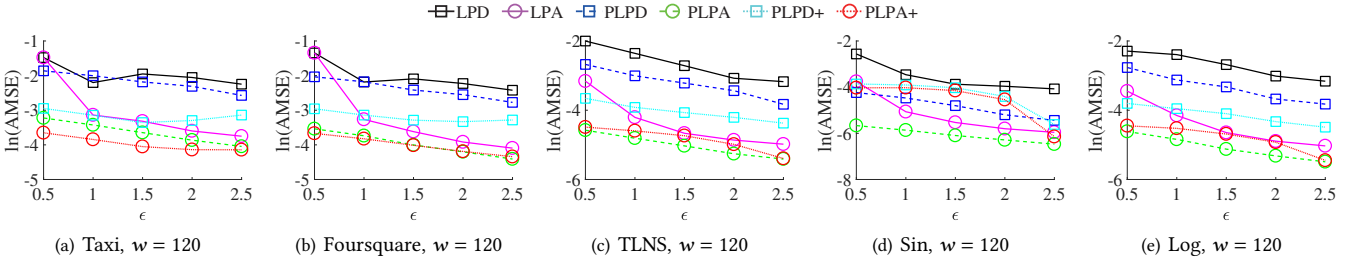


Figure 4: AMSE with different ϵ .

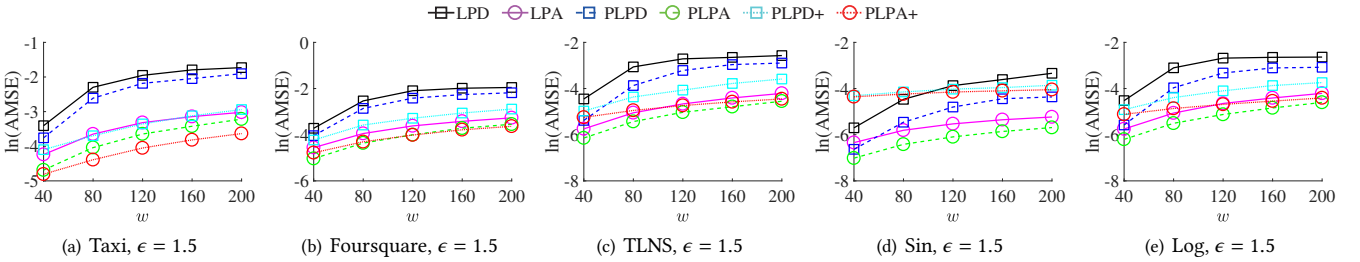


Figure 5: AMSE with different w .

6.3 Time Cost Analysis

Figure 2 illustrates the running time as the privacy budget ϵ increases. Overall, our PLPD, PLPA, PLPD+ and PLPA+ incur slightly higher running time than non-personalized methods due to the additional overhead of window size selection and user grouping. As ϵ increases, the running time of all methods except PLPD+ and PLPA+ remains nearly constant. This is because, for non-personalized methods and baseline personalized methods, the privacy budget has little impact on the computational complexity. In contrast, the running time of both PLPD+ and PLPA+ decreases as ϵ increases. This is

because a larger ϵ leads to fewer groups in our experimental parameter settings, which further reduces the computing group statistics in OPS and PUE.

Figure 3 shows the running time under different the window sizes w . In general, our PLPD, PLPA, PLPD+ and PLPA+ exhibit slightly higher running time than non-personalized methods, owing to the extra cost introduced by window size selection and user grouping. As w increases, the running time of all methods except PLPD+ and PLPA+ decreases initially and then levels off. This trend arises because a larger window size leads to a smaller sampling rate, thereby reducing the amount of statistic processing. Conversely,

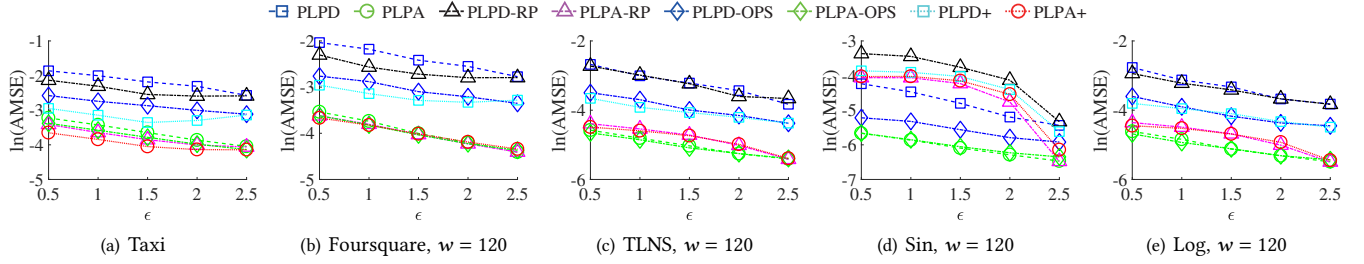


Figure 6: The ablation study of AMSE with different ϵ .

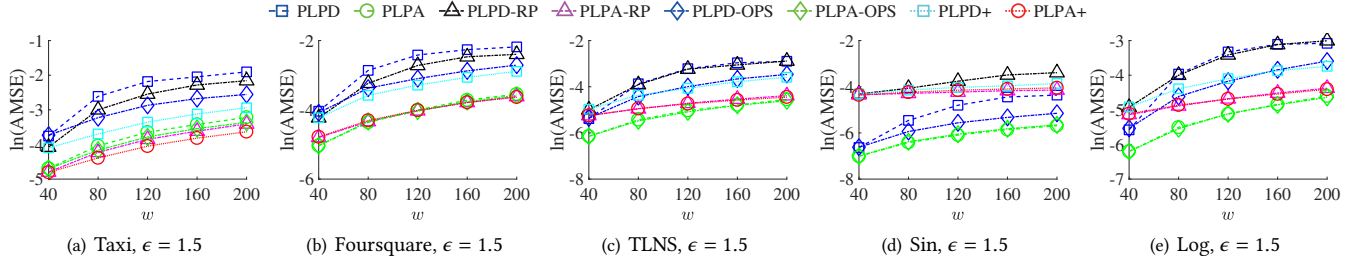


Figure 7: The ablation study of AMSE with different w .

PLPD⁺ and PLPA⁺ experience increasing execution time with larger w . Given our experimental configuration, enlarging w produces a greater number of user groups, which in turn raises the overhead of group-wise statistics computation in OPS and PUE.

6.4 Overall Utility Analysis

Figure 4 presents the publication accuracy of all methods as the privacy budget ϵ varies. In general, the logarithmic error $\ln(AMSE)$ of all methods decreases as ϵ increases, reflecting the trade off between privacy and data utility. The absorption-based methods (i.e., LBA, PLPA, PLPA⁺) consistently outperform their corresponding distribution-based counterparts (i.e., LBD, PLPD, PLPD⁺). This is because the distribution-based methods distribute population in an exponentially decaying way in the window, leading to quite larger estimation error. More over, our personalized mechanisms outperform the corresponding non-personalized ones in most datasets as ϵ increases, demonstrating the benefits of personalized methods. Additionally, our PLPA⁺ yields the best overall performance on real datasets, with an average error 47.7% lower than LBA, while our PLPA performs best on synthetic datasets, reducing the average error by 47.27% compared with LBA. This difference arises because when the data dimension d is small, the advantages of OPS and PUE become less pronounced, leading PLPA to outperform PLPA⁺ in low-dimensional synthetic datasets.

Figure 5 shows the publication accuracy as the window size w increases. Overall, $\ln(AMSE)$ increases with larger w , since a wider window reduces the number of users contributing to each time slot. Similar to the previous trend, absorption-based methods (i.e., LBA, PLPA, PLPA⁺) still outperform distribution-based ones (i.e., LBD, PLPD, PLPD⁺) for the same reason—exponentially decaying population assignment results in higher estimation error. More over, our personalized mechanisms outperform their non-personalized counterparts, validating the benefit of personalization. Consistent with the results on ϵ , our PLPA⁺ achieves the best accuracy on real

datasets, with an average error 38.65% lower than LBA, whereas our PLPA excels on synthetic datasets, reducing the average error by 36.5% compared with LBA.

6.5 Ablation Study

To evaluate the contributions of our sub-modules OPS and PUE, we conduct an ablation study by selectively removing them. We construct four ablated variants: PLBD-RP, PLBA-RP, PLBD-OPS and PLBA-OPS. Specifically, PLBD-RP and PLBA-RP are the enhanced population methods without OPS, while PLBD-OPS and PLBA-OPS are the enhanced population methods without PUE.

Figure 6 and Figure 7 compare the accuracy of these ablated methods with our complete solutions as ϵ and w vary, respectively. As shown, all methods exhibit similar trends: the error decreases with larger ϵ and increase with larger w . Besides, the inclusion of OPS yields substantial accuracy gains over the basic population methods in most cases and maintains comparable performance otherwise. Similarly, PUE also improves accuracy, especially on real datasets where heterogeneity among users is more evident. When the data dimension decreases to $d = 2$, however, the advantage of PUE diminishes or even reverses—particularly in datasets such as *Sin*, where the underlying data exhibit non-stationary variation rates.

7 CONCLUSION

In this paper, we propose two basic methods PLPD and PLPA for Personalized Private Steaming Data Estimation in Local Setting. To further improve the utility, we propose two enhanced methods, PLPD⁺ and PLPA⁺. We evaluate these methods against recent non-personalized approaches, to demonstrate their efficiency and effectiveness.

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8 APPENDIX

8.1 Process of PLPD and PLPA

We introduce the process of PLPD and PLPA for subsection 4.3 as follows.

PLDP Population Distribution. PLPD takes an available user set U_A , the privacy requirement (w, ϵ) of all users, the value domain d and historical publications as input. PLPD starts with a total

Algorithm 5: PLDP Population Distribution

Input: Available user set U_A , privacy requirement (w, ϵ) of all Users U , dataset D_t data domain size d , historical data publication $(\hat{f}_1, \hat{f}_2, \dots, \hat{f}_{t-1})$

Output: $\hat{f}_t$

- 1 Calculate the distinct privacy budget $\tilde{\epsilon}$ with the statistic g ;
- 2 Get the maximal window size $w_{\max} = \max(w)$;
- 3 // sub-mechanism $M_{s,t}$
- 3 Calculate the minimal sampling size $z_{\min} \leftarrow \lfloor n/(2w_{\max}) \rfloor$;
- 4 Sample users $U_{s,t}$ from U_A with the size of $z_{\min}$ and remove $U_{s,t}$ from U_A , i.e., $U_A = U_A \setminus U_{s,t}$;
- 5 Get report $Y_t \leftarrow \text{PFO.P}(D_t)$ with privacy budget list ϵ from users in $U_{s,t}$;
- 6 Get the combination $h_{s,t} \leftarrow \text{PFO.A}(Y_t)$;
- 7 Estimate $\hat{f}_{s,t} \leftarrow \text{PFO.E}(h_{s,t}, \epsilon)$;
- 8 Calculate $dis \leftarrow \frac{1}{d} \sum_{j=1}^d (\hat{f}_{s,t}[j] - \hat{f}_t[j])^2 - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]]$;
- 9 // sub-mechanism $M_{r,t}$
- 9 Calculate the minimal remaining population size $n_{r,\min} = n/2 - \sum_{\tau=t-w_{\max}+1}^{t-1} |U_{r,\tau}|$;
- 10 Set the minimal number of potential publication users $n_{pp,\min} = \lfloor n_{r,\min}/2 \rfloor$;
- 11 Calculate the potential reporting error $err \leftarrow err(\hat{\epsilon}, g, n, n_{pp,\min}, d)$ by Equation (7);
- 12 **if** $dis > err$ **and** $n_{pp,\min} \geq n_{\min}$ **then**
- 13 Sample a user set $U_{r,t}$ from U_A with the size of $|U_{r,t}| = n_{pp,\min}$ and remove $U_{r,t}$ from U_A , i.e., $U_A \leftarrow U_A \setminus U_{r,t}$;
- 14 $h_{r,t} \leftarrow$ Users in $U_{r,t}$ report via an PFO.P with privacy budget list ϵ ;
- 15 Calculate $\hat{f}_{r,t} \leftarrow \text{PFO.E}(h_{r,t}, \epsilon)$;
- 16 **else**
- 17 Set $\hat{f}_{r,t} \leftarrow \hat{f}_{r,t-1}$;
- 18 **if** $t \geq w_{\max}$ **then**
- 19 $U_A \leftarrow U_A \cup U_{s,t-w_{\max}+1} \cup U_{r,t-w_{\max}+1}$;
- 20 **return** $\hat{f}_{r,t}$.

population ($U_A = U$) and executes three steps: (1) calculates a fixed population per time slot; (2) distributes publication population in an exponential decreasing fashion; (3) recycles the population spent in time slots falling outside the activate window. Algorithm 5 shows the details of PLPD.

PLPD first gets different privacy budgets' statistic which is used for PFO. It finds the max window size $w_{\max}$ and calculate the minimal population average share $z_{\min}$ which ensures the total population spent within any window is not large than the total population U . PLPD samples $z_{\min}$ size of population $U_{s,t}$ (Line 4) and executes PFO using $U_{s,t}$ (Line 5 and Line 7) to get the estimation frequency of current data. It then uses these estimation to calculate the private dissimilarity (Line 8). Rather than directly calculating new reporting by spending half of the remaining population, PLPD first calculates the new reporting error without any population spent (Lines 9–11). It judges whether the dissimilarity is larger than the error. If so, it spends the population and reports new estimation (Lines 14–15). Otherwise, it sets the current estimation as the before estimation. Finally, PLPD recycles the population that spent for private dissimilarity calculation and new estimation at time slot

$t - w_{\max} + 1$. We give an example for the procedure of PLPD in Example 6.

Example 6. Consider a scenario with 5 two-by-two connected locations $\{A, B, C, D, E\}$ and 1000 users $u_1, u_2, \dots, u_{1000}$. These users have privacy requirements in domain $\{0.2, 0.4, 0.6, 0.8\}$ and window size requirements in domain $\{1, 2, 3, 4\}$. From this, we can determine that the input and output domain size is $d = 5$. The distinct privacy budget vector is $\tilde{\epsilon} = \langle 0.2, 0.4, 0.6, 0.8 \rangle$, with probability vectors $\tilde{q} = \langle 0.19, 0.18, 0.17, 0.16 \rangle$ and $\tilde{p} = \langle 0.23, 0.27, 0.31, 0.36 \rangle$. The corresponding statistic vector of $\tilde{\epsilon}$ is $g = \langle 0.35, 0.25, 0.25, 0.15 \rangle$. The maximum window size is $w_{\max} = 4$, and the minimum sampling size is $z_{\min} = \lfloor 1000/(2 \times 4) \rfloor = 125$.

At time slot 1, the available user set U_A is set to U . It samples 125 users. Assume these 125 users are $U_{s,1} = \{u_1, u_2, \dots, u_{125}\}$. Then U_A is updated as $U_A = U_A \setminus U_{s,1} = \{u_{126}, u_{127}, \dots, u_{1000}\}$. Users in $U_{s,1}$ report their obfuscated locations using PFO.P. Assume the obfuscated frequency is $h_{s,1} = \langle 0.22, 0.2, 0.18, 0.22, 0.18 \rangle$. We can then obtain the estimation by Equation (1) as $\hat{f}_{s,1} = \langle 0.396, 0.2, 0.004, 0.396, 0.004 \rangle$ with variance sum as $\sum_{j=1}^d \text{Var}[\hat{f}_{s,1}[j]] = 0.606$. Next, we calculate the dissimilarity $dis = 0.185$. The potential publication user number $n_{pp,\min} = \lfloor \frac{n}{2} \rfloor = 250$. Thus, we can calculate $err = 0.061$. Since $dis > err$, the system continues to sample 250 users from U_A for a new publication. Assume the sampling set is $U_{r,1} = \{u_{126}, u_{127}, \dots, u_{375}\}$. These users report their obfuscated locations using PFO.P. Assume the obfuscated frequency is $h_{r,1} = \langle 0.23, 0.19, 0.19, 0.21, 0.18 \rangle$. We can then get the estimation as $\hat{f}_{r,1} = \langle 0.494, 0.102, 0.102, 0.300, 0.04 \rangle$. Because there are no users for historical publications out of window size $w_{\max}$ (i.e., $t < w_{\max}$), U_A remain unchanged.

At time slot 2, it still samples 125 users from U_A for dis calculation. Assume these 125 users are $U_{s,2} = \{u_{376}, u_{377}, \dots, u_{500}\}$. Then the available user set is updated as $U_A = U_A \setminus U_{s,2} = \{u_{501}, u_{502}, \dots, u_{1000}\}$. The system still calculates the minimum remaining population size $n_{r,\min} = 1000/2 - 250 = 250$. Then, it calculates the potential publication user number $n_{pp,\min} = \lfloor n_{r,\min}/2 \rfloor = 125$ and err . Assume $dis > err$, it continues to sample $n_{pp,\min} = 125$ users for a new publication. Assume the sampling user set is $U_{r,2} = \{u_{501}, u_{502}, \dots, u_{625}\}$. Then, the available user set is updated as $U_A = U_A \setminus U_{r,2} = \{u_{626}, u_{627}, \dots, u_{1000}\}$. It then reports a new publication $\hat{f}_{r,2}$.

Assume there are no new publications from time slot 3 (i.e., $dis \leq err$ and $\hat{f}_{r,3} = \hat{f}_{r,2}$). The system still consumes $z_{\min} = 125$ population for dissimilarity calculation. Assume the consumed population set is $U_{r,3} = \{u_{626}, u_{627}, \dots, u_{750}\}$. At time slot 4, the available user set is $U_A = \{u_{751}, u_{752}, \dots, u_{1000}\}$. The system samples $z_{\min} = 125$ users for dissimilarity calculation. Assume the sampling user set is $U_{s,4} = \{u_{751}, u_{752}, \dots, u_{875}\}$. Then U_A is updated as $U_A = \{u_{876}, u_{877}, \dots, u_{1000}\}$. Assume $dis > err$. The system calculates the minimum remaining population size $n_{r,\min} = 1000/2 - (250 + 125) = 125$ and samples $n_{pp,\min} = \lfloor n_{pp,\min}/2 \rfloor = 62$ users for new publication. Assume these 62 users are $U_{r,4} = \{u_{876}, u_{877}, \dots, u_{937}\}$. It then reports a new publication $\hat{f}_{r,4}$. Additionally, it recycles the population consumption from time slot 1, specifically, $U_A = U_A \cup U_{s,1} \cup U_{r,1} = \{u_1, u_2, \dots, u_{375}, u_{938}, u_{939}, \dots, u_{1000}\}$.

The process repeats in a similar fashion for subsequent time slots.

PLDP Population Absorption. PLPD performs well for steams with frequent but gradual changes. However, it becomes less effective when dealing with steams that changes that change infrequently but dramatically. To address this scenario, we propose PLDP Population Absorption (PLPA).

Algorithm 6 shows the process of PLPA. In PLPA, like PLPD, it calculates the statistics $\mathbf{g}$ of distinct privacy budget $\tilde{\epsilon}$ and the maximum window size w_{max} among all users. The process of dissimilarity calculation remains the same as in PLPD. However, the population allocation method is different. In PLPA, it first pre-allocates to $\lfloor n/(2w_{max}) \rfloor$ population at each time slot. This population value, called unit population, is the smallest allocation and cannot be further divided. Each unit population can be nullified by predecessors or absorbed by successors within the w_{max} window. However, the average population consumption across each w_{max} period is not allowed to exceed the unit population.

Especially, for error calculation, the system first determines the current nullified t_N , which represents the number of consecutive time slots—from $l+1$ to $l+t_N$ —whose populations are nullified by the last new publication time slot l . These t_N population resources are marked as null. PLPA checks whether the current time slot t is before $l+t_N$ (Line 5). If so, the population at time slot t is remains nullified and needs to be skipped (Line 6). Otherwise, it calculates the number of time slot t_A that can be absorbed from $l+t_N+1$ to t (Line 8) and absorbs them to increase the current population (Line 9). The error calculation (Line 10) and dissimilarity-error comparison process (Lines 11–16) follow the same approach as in PLPD. Finally, PLPA recycles the historical population at time slot $t-w_{max}+1$.

We give an example for PLPA as follows.

Example 7. Assume the users are the same in Example 6. At time slot 1, PLPA spends $U_{s,1} = \{u_1, u_2, \dots, u_{125}\}$ in calculating the dissimilarity $dis = 0.185$. It then gets the user set $|U_{r,1}| = 0$ consumption for publication at the last time $l = 0$ and calculates the nullified $t_N = -1$. Because $t - l = 1 > t_N$, PLPA calculates $t_A = t - l - t_N = 2$, which mean the new publication can consume two shares of populations. It calculate the two share population quantity $n_{pp,min} = \lfloor 1000/(2 \times 4) \rfloor \times 2 = 250$ and the potential reporting error $err = 0.03$. Because $dis > err$, PLPA samples a user set $|U_{r,1}|$ of size 250 from U_A . Assume $|U_{r,1}| = \{u_{126}, u_{127}, \dots, u_{375}\}$. PLPA reports a new publication consuming $|U_{r,1}|$.

At time slot 1, assume the sampling 125 users for dissimilarity calculation is $U_{s,2} = \{u_{376}, u_{377}, \dots, u_{500}\}$. Thus the available user set is updated as $U_A = U_A \setminus U_{s,2} = \{u_{500}, u_{501}, \dots, u_{1000}\}$. The nullified $t_N = \frac{250}{1000/(2 \times 4)} = 2$, which is larger than $t - l = 1$. Thus, PLPA approximates the current publication $\hat{f}_2$ as the last publication $\hat{f}_1$.

The process repeats in a similar way for subsequent time slots.

8.2 More Details for Modified Personalized Frequency Oracle Theorems

Theorem 8.1. GPRR is Φ -PLDP and $\hat{f}_j = \sum_{k=1}^m \alpha_k \cdot \frac{h_{k,j} - \tilde{q}_k}{\tilde{p}_k - \tilde{q}_k}$ is an unbiased aggregation of f_j for all $\omega_j \in \Omega$.

Algorithm 6: PLDP Population Absorption

Input: Available user set U_A , privacy requirement $(\mathbf{w}, \epsilon)$ of all Users U , data domain size d , historical data publication $(\hat{f}_1, \hat{f}_2, \dots, \hat{f}_{t-1})$

Output: $\hat{f}_t$

- 1 Calculate the distinct privacy budget $\tilde{\epsilon}$ with the statistic $\mathbf{g}$;
- 2 Get the maximal window size $w_{max} = \max(\mathbf{w})$;
- 3 // sub-mechanism $M_{s,t}$
- 3 The same as Line 3-8 in Algorithm 5;
- 3 // sub-mechanism $M_{r,t}$
- 4 Calculate the nullified $t_N = \lfloor \frac{|U_{r,l}|}{\lfloor n/(2w_{max}) \rfloor} \rfloor - 1$;
- 5 **if** $t - l < t_N$ **then**
- 6 $\hat{f}_t \leftarrow \hat{f}_{t-1}$;
- 7 **else**
- 8 Calculate time slots that can be absorbed $t_A = t - l - t_N$;
- 9 Set the number of potential publication users $n_{pp,min} \leftarrow \lfloor n/(2w_{max}) \rfloor \cdot \min(t_A, w_{max})$;
- 10 Calculate the potential reporting error err by Equation (7);
- 11 **if** $dis > err$ **then**
- 12 Sample a user set $U_{r,t}$ from U_A with the size of $|U_{r,t}| = n_{pp,min}$ and remove $U_{r,t}$ from U_A , i.e., $U_A \leftarrow U_A \setminus U_{r,t}$;
- 13 $\mathbf{h}_{r,t} \leftarrow$ Users in $U_{r,t}$ report via an PFO.P with privacy budget list ϵ ;
- 14 Calculate $\hat{f}_{r,t} \leftarrow \text{PFO.E}(\mathbf{h}_{r,t}, \epsilon)$;
- 15 **else**
- 16 Set $\hat{f}_{r,t} \leftarrow \hat{f}_{r,t-1}$;
- 17 **if** $t \geq w_{max}$ **then**
- 18 $U_A \leftarrow U_A \cup U_{s,t-w_{max}+1} \cup U_{r,t-w_{max}+1}$;
- 19 **return** $\hat{f}_{r,t}$.

PROOF. For any two input $x_i, x'_i \in \Omega$ of u_i with the output $y_i \in \Omega$, we have

$$\frac{\Pr[M(x_i) = y_i]}{\Pr[M(x'_i) = y_i]} \leq \frac{p_i}{q_i} = e^{\epsilon_i}. \quad (9)$$

Thus, GPRR is Φ -PLDP.

For any expectation $\hat{f}_j \in \hat{\mathbf{f}}$ of $f_j \in \mathbf{f}$, we have

$$\begin{aligned} \mathbb{E}[\hat{f}_j] &= \mathbb{E}\left[\sum_{k=1}^m \alpha_k \cdot \frac{h_{k,j} - \tilde{q}_k}{\tilde{p}_k - \tilde{q}_k}\right] \\ &= \mathbb{E}\left[\sum_{k=1}^m \alpha_k \cdot \frac{f_j \cdot \tilde{p}_k + (1 - f_j) \tilde{q}_k - \tilde{q}_k}{\tilde{p}_k - \tilde{q}_k}\right] \\ &= \mathbb{E}\left[f_j \cdot \sum_{k=1}^m \alpha_k\right] \\ &= f_j. \end{aligned}$$

Thus, $\hat{f}_j$ is an unbiased estimation of f_j . $\square$

Theorem 8.2. For any Φ -PLDP mechanism with $\Phi : U \rightarrow \mathcal{E}$ independent of Ω , and any $\tilde{p}_k \in \tilde{\mathbf{p}}, \tilde{q}_k \in \tilde{\mathbf{q}}$ with $\frac{1}{d} < \tilde{p}_k \leq 1$ and $\tilde{q}_k = \frac{1 - \tilde{p}_k}{d-1}$, the variance of the j -th frequency estimation $\hat{f}_j \in \hat{\mathbf{f}}$ in

Equation (1) is

$$\text{Var}[\hat{f}_j] = \begin{cases} V_{A,j} = \frac{1}{\sum_{k=1}^m 1/(\lambda_k + \mu_k \cdot f_j)}, & \text{if } \alpha_k = \frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\text{Var}[\hat{f}_{k',j}]}, \\ V_{B,j} = \frac{\sum_{k=1}^m \frac{(\lambda_k + \mu_k/d)}{(\lambda_k + \mu_k \cdot f_j)^2}}{\left(\sum_{k=1}^m 1/(\lambda_k + \mu_k/d)\right)^2}, & \text{if } \alpha_k = \frac{1/\text{Var}_a[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\text{Var}_a[\hat{f}_{k',j}]}; \end{cases} \quad (10)$$

Besides, the upper bound of $\sum_{j=1}^d V_{A,j}$ and the lower bound of $\sum_{j=1}^d V_{B,j}$ are both

$$\text{Var}[\hat{f}] = \frac{1}{\sum_{k=1}^m n/(d\lambda_k + \mu_k)}, \quad (11)$$

where $\lambda_k = \frac{\tilde{q}_k(1-\tilde{q}_k)}{g_k(\tilde{p}_k - \tilde{q}_k)^2}$ and $\mu_k = \frac{1-\tilde{p}_k - \tilde{q}_k}{g_k(\tilde{p}_k - \tilde{q}_k)}$.

PROOF. For any random variable $\hat{h}_j \in \mathbf{h}$, it is the average of n independent random variables drawn from the Bernoulli distribution. More specifically, $f_j g_k n$ of these random variables are drawn from the Bernoulli distribution with parameter $\tilde{p}_k$. Besides, $(1 - g_k) f_j n$ of these variables are drawn from the Bernoulli distribution with parameter $\tilde{q}_k$. Thus, if setting $\alpha_k = \frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\text{Var}[\hat{f}_{k',j}]}$, then

$$\begin{aligned} \text{Var}[\hat{f}_j] &= \sum_{k=1}^m \alpha_k^2 \cdot \text{Var}[\hat{f}_{k,j}] \\ &= \sum_{k=1}^m \left(\frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{i=1}^m 1/\text{Var}[\hat{f}_{i,j}]} \right)^2 \cdot \text{Var}[\hat{f}_{k,j}] \\ &= \frac{\sum_{k=1}^m 1/\text{Var}[\hat{f}_{k,j}]}{\left(\sum_{i=1}^m 1/\text{Var}[\hat{f}_{i,j}]\right)^2} \\ &= \frac{1}{\sum_{k=1}^m 1/\text{Var}[\hat{f}_{k,j}]} \\ &= 1/\sum_{k=1}^m 1/(\lambda_k + \mu_k \cdot f_j); \end{aligned} \quad (12)$$

If setting $\alpha_k = \frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{k'=1}^m 1/\text{Var}[\hat{f}_{k',j}]}$, then

$$\begin{aligned} \text{Var}[\hat{f}_j] &= \sum_{k=1}^m \alpha_k^2 \cdot \text{Var}[\hat{f}_{k,j}] \\ &= \sum_{k=1}^m \left(\frac{1/\text{Var}[\hat{f}_{k,j}]}{\sum_{i=1}^m 1/\text{Var}[\hat{f}_{i,j}]} \right)^2 \cdot \text{Var}[\hat{f}_{k,j}] \\ &= \frac{\sum_{k=1}^m \text{Var}[\hat{f}_{k,j}] / (\text{Var}[\hat{f}_{k,j}])^2}{\left(\sum_{i=1}^m 1/\text{Var}[\hat{f}_{i,j}]\right)^2} \\ &= \frac{\sum_{k=1}^m (\lambda_k + \mu_k/d) / (\lambda_k + \mu_k \cdot f_j)^2}{\left(\sum_{k=1}^m 1/(\lambda_k + \mu_k/d)\right)^2} \end{aligned} \quad (13)$$

For Equation (12), let $\phi(x) = \text{Var}[x] = \frac{1}{\sum_k \frac{1}{\lambda_k + \mu_k x}}$, then we have:

$$\sum_{j=1}^d V_{A,j} = \sum_{j=1}^d \phi(\hat{f}_j) = \frac{1}{m} H(\lambda_1 + \mu_1 f_j, \lambda_2 + \mu_2 f_j, \dots, \lambda_m + \mu_m f_j),$$

where H is the harmonic average. $\square$

From Equation (10) and (11), we can see the upper bound of $\sum_{j=1}^d V_{A,j}$ is independent of f_j .

Because $\lambda_k + \mu_k f_j > 0$, $\sum_{j=1}^d V_{A,j}$ is a concave function. As $\sum_j f_j = 1$, according to Jensen's inequality, we have:

$$\sum_{j=1}^d V_{A,j} = \sum_{j=1}^d \phi(\hat{f}_j) \leq d \cdot \phi\left(\frac{1}{d}\right) = d \cdot \frac{1}{\sum_{k=1}^m \frac{1}{\lambda_k + \mu_k/d}} = \frac{1}{\sum_{k=1}^m \frac{1}{d\lambda_k + \mu_k}}.$$

The equal sign holds when $f_1 = f_2 = \dots = f_d = \frac{1}{d}$.

For Equation (13), let $\psi(x) = \sum_{k=1}^m \frac{\lambda_k + \mu_k/d}{(\lambda_k + \mu_k f_j)^2}$, then we have:

$$\sum_{j=1}^d V_{B,j} = \sum_{j=1}^d \psi(\hat{f}_j) = \frac{1}{\left(\sum_{k=1}^m \frac{1}{\lambda_k + \mu_k/d}\right)^2} \sum_{j=1}^d \sum_{k=1}^m \frac{\lambda_k + \mu_k/d}{(\lambda_k + \mu_k f_j)^2}.$$

Because $\psi(x)$ is a decreasing convex function, according to Jensen's inequality, we have

$$\sum_{j=1}^d V_{B,j} = \sum_{j=1}^d \psi(f_j) \geq d \cdot \psi\left(\frac{1}{d}\right) = d \cdot \frac{1}{\sum_{k=1}^m \frac{1}{\lambda_k + \mu_k/d}} = \frac{1}{\sum_{k=1}^m \frac{1}{d\lambda_k + \mu_k}}.$$

The equal sign holds when $f_1 = f_2 = \dots = f_d = \frac{1}{d}$.

8.3 More Details for Dissimilarity

Private Dissimilarity. The ideal method of strategy determination is using non-private dissimilarity dis^* for comparison. The expression of non-private dissimilarity is defined as the mean square error between the

$$dis^* = \frac{1}{d} \sum_{j=1}^d (f_t[j] - \hat{f}_l[j])^2, \quad (14)$$

where $\hat{f}_l$ represents the previous frequency publication that consumes population resources. However, dis^* as an intermediate result leads to privacy leakage. To overcome this limitation, we use a private dissimilarity dis instead. The design of dis has two requirements: (1) dis needs to be private, meaning that given a privacy requirement Φ for dis calculation, it needs to achieve Φ -PLDP; (2) dis needs to be an unbiased estimation of the non-private dissimilarity dis^* , ensuring accurate results.

Theorem 8.3. Let $\hat{f}_{s,t}$ denote the unbiased frequency estimation of f_t from an Φ -PLDP frequency estimation mechanism $M_{s,t}$ at current time slot t . Then the following dissimilarity measure

$$dis = \frac{1}{d} \sum_{j=1}^d (\hat{f}_{s,t}[j] - \hat{f}_l[j])^2 - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]] \quad (15)$$

is Φ -PLDP and an unbiased estimation of dis^* in Equation (14).

PROOF. We first prove (1) dis in Equation (15) is Φ -PLDP (Privacy Proof) and then prove (2) dis in Equation (15) is an unbiased estimation of dis^* in Equation (14) (Unbiasedness Proof).

Privacy Proof. The calculation of dis is only related to $\{y_1, \dots, y_n\}$ of all users, which is published utilizing GRR. Thus, According to Theorem 8.1, we get dis is Φ -PLDP.

Unbiasedness Proof. Since $\hat{f}_{s,t}$ is an unbiased estimation of f_t , for any $\hat{f}_{s,t}[j] \in \hat{f}_{s,t}$ and $f_t[j] \in f_t$, we have

$$\mathbb{E}[\hat{f}_{s,t}[j]] = f_t[j].$$

Besides, for any random variable X and constant C , according to $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, we have

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[(X - C) - (\mathbb{E}[X] - C)]^2 \\ &= \mathbb{E}[(X - C)^2] - (\mathbb{E}[X] - C)^2. \end{aligned}$$

Since last reporting value $\hat{f}_l$ has been a constant for the current value $\hat{f}_{s,t}$, thus, we have

$$\text{Var}[\hat{f}_{s,t}[j]] = \mathbb{E}[(\hat{f}_{s,t}[j] - \hat{f}_l[j])^2] - (f_t[j] - \hat{f}_l[j])^2.$$

Therefore,

$$\mathbb{E}[(\hat{f}_{s,t}[j] - \hat{f}_l[j])^2] = (f_t[j] - \hat{f}_l[j])^2 + \text{Var}[\hat{f}_{s,t}[j]].$$

Hence, the expectation of dis in Equation (15) satisfies

$$\begin{aligned} \mathbb{E}[dis] &= \mathbb{E}\left[\frac{1}{d} \sum_{j=1}^d (\hat{f}_{s,t}[j] - \hat{f}_l[j])^2 - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]]\right] \\ &= \frac{1}{d} \sum_{j=1}^d \mathbb{E}[(\hat{f}_{s,t}[j] - \hat{f}_l[j])^2] - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]] \\ &= \frac{1}{d} \sum_{j=1}^d \left((f_t[j] - \hat{f}_l[j])^2 + \text{Var}[\hat{f}_{s,t}[j]] \right) - \frac{1}{d} \sum_{j=1}^d \text{Var}[\hat{f}_{s,t}[j]] \\ &= \frac{1}{d} \sum_{j=1}^d (f_t[j] - \hat{f}_l[j])^2 \\ &= dis^*. \end{aligned}$$

□

8.4 Proof of the Theorem for Re-Perturbation (Theorem 5.1)

PROOF. Let FP_H and FP_L be the sub-mechanisms in GPRR with privacy budget ϵ_H and ϵ_L , respectively. Then, we have

$$\forall y \in Y, \Pr[\text{FP}_H(x) = y] = \begin{cases} p_H, & \text{if } y = x, \\ q_H = \frac{1-p_H}{d-1}, & \text{otherwise,} \end{cases}$$

and

$$\forall y \in Y, \Pr[\text{FP}_L(x) = y] = \begin{cases} p_L, & \text{if } y = x, \\ q_L = \frac{1-p_L}{d-1}, & \text{otherwise.} \end{cases}$$

For any re-perturbed value Y' , we have the transformation probability relationship shown in Figure 8.

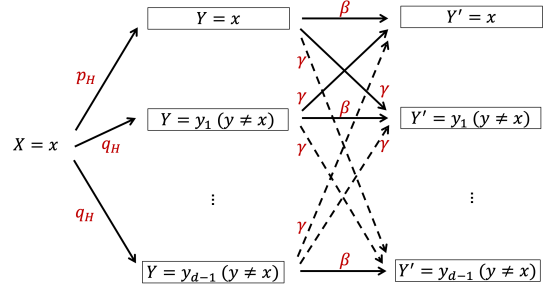


Figure 8: Probability for perturbation and re-perturbation.

Thus, $\forall y'_1, y'_2 \in Y'$, we have:

$$\begin{aligned} &\frac{\Pr[M(x) = y'_1]}{\Pr[M(x) = y'_2]} \\ &= \frac{\sum_{y \in Y} \Pr[\text{FP}(x) = y] \cdot \Pr[\text{RP}(y) = y'_1]}{\sum_{y \in Y} \Pr[\text{FP}(x) = y] \cdot \Pr[\text{RP}(y) = y'_2]} \\ &\leq \frac{p_H \cdot \beta + (d-1) \cdot q_H \cdot \gamma}{p_H \cdot \gamma + q_H \cdot \beta + (d-2) \cdot q_H \cdot \gamma} \\ &= \frac{p_H \cdot \frac{dp_L - p_L + p_H - 1}{d-1} + (d-1) \cdot \frac{1-p_H}{d-1} \cdot \frac{p_H - p_L}{dp_H - 1}}{p_H \cdot \frac{p_H - p_L}{dp_H - 1} + \frac{1-p_H}{d-1} \cdot \frac{dp_L - p_L + p_H - 1}{dp_H - 1} + (d-2) \cdot \frac{1-p_H}{d-1} \cdot \frac{p_H - p_L}{dp_H - 1}} \\ &= \frac{(d-1)(dp_H p_L - p_L)}{(p_H + d-2)(p_H - p_L) + (1-p_H)(dp_L - p_L + p_H - 1)} \\ &= \frac{(d-1)p_L(dp_H - 1)}{(dp_H - 1)(1 - p_L)} \\ &= \frac{p_L}{\frac{1-p_L}{d-1}} \\ &= \frac{p_L}{q_L} \\ &= e^{\epsilon_L}. \end{aligned}$$

Therefore, after the re-perturbing process, it still satisfies ϵ_L -LDP.

Noted that FP_L satisfies ϵ_L -LDP. According to Reference [36], we can calculate FP_L 's variance with z_L users is

$$\begin{aligned} \text{Var}_L(z_L) &= \frac{1}{z_L^2} \cdot \sum_{j=1}^d \left(\frac{z_L q_L (1 - q_L)}{(p_L - q_L)^2} + \frac{z_L (1 - p_L - q_L)}{p_L - q_L} \cdot f_j \right) \\ &= \frac{dq_L (1 - q_L)}{z_L (p_L - q_L)^2} + \frac{1 - p_L - q_L}{z_L (p_L - q_L)} \end{aligned}$$

After re-perturbing, the variance becomes

$$\frac{dq_L (1 - q_L)}{(z_L + z_H) (p_L - q_L)^2} + \frac{1 - p_L - q_L}{(z_L + z_H) (p_L - q_L)}$$

Therefore the improvement is

$$\begin{aligned} \Delta(\text{Var}) &= \text{Var}_L(z_L) - \text{Var}_L(z_L + z_H) \\ &= \frac{z_H}{z_L (z_L + z_H)} \left(\frac{dq_L (1 - q_L)}{(p_L - q_L)^2} + \frac{1 - p_L - q_L}{p_L - q_L} \right). \end{aligned}$$

□

8.5 Proofs of Theorems for Time Complexity

8.5.1 Proof of Theorems 4.1.

PROOF. The time complexities of PFO.P, PFO.A and PFO.E are $O(z_{\min})$, $O(z_{\min} + d \cdot m)$ and $O(d \cdot m)$, respectively. Additionally, the time complexity of dis calculation is $O(d)$. Therefore, the overall

time complexity of sub-mechanism $\mathcal{M}_{s,t}$ is $O(z_{\min} + d \cdot m)$ for both PLPD and PLPA.

For sub-mechanism $\mathcal{M}_{r,t}$ in PLPD, the sampling size is at most $n/4$. Thus, the time complexity for counting distinct elements in $\tilde{\epsilon}$ is $O(n)$. The time complexity of the FO is $O(n + d \cdot m)$. Thus, the time complexity of $\mathcal{M}_{r,t}$ in PLPD is $O(n + d \cdot m)$. For sub-mechanism $\mathcal{M}_{r,t}$ in PLPA, the sampling size is at most $\max(\lfloor n/(2w_{\max}) \rfloor) \cdot w_{\max} = n/2$. Hence, the time complexity of $\mathcal{M}_{r,t}$ in PLPA is also $O(n + d \cdot m)$.

Consequently, the time complexity for both PLPD and PLPA is $O(n + d \cdot m)$. $\square$

8.5.2 Proof of Theorems 5.2.

PROOF. The time complexity of OPS is $O(m \cdot n)$ in both PLPD⁺ and PLPA⁺. Since OPS is executed only once at the beginning of the process, its impact is minimal in subsequent steps.

The time complexity of PUE is $O(m^2)$. After applying PUE, the sampling size increases from z to at most $z \cdot m$. The time complexities of PFO.P, PFO.A and PFO.E are $O(z_{\text{opt}})$, $O(z_{\text{opt}} \cdot m + d \cdot m)$ and $O(d \cdot m)$, respectively. Additionally, the time complexity of *dis* calculation is $O(d)$. Therefore, the overall time complexity of sub-mechanism $\mathcal{M}_{s,t}$ is $O(m^2 + z_{\text{opt}} \cdot m + d \cdot m)$ for both PLPD⁺ and PLPA⁺.

For sub-mechanism $\mathcal{M}_{r,t}$ in PLPD⁺, the sampling size is at most $n/4$. Thus, the time complexity for counting distinct elements in $\tilde{\epsilon}_{\text{opt}}$ is $O(n)$. The time complexity of the enhanced PFO is $O(n \cdot m + d \cdot m)$. Thus, the time complexity of $\mathcal{M}_{r,t}$ in PLPD⁺ is $O(n \cdot m + d \cdot m)$. For sub-mechanism $\mathcal{M}_{r,t}$ in PLPA⁺, the sampling size is at most $\max(\lfloor n/(2w_{\text{opt}}) \rfloor) \cdot w_{\text{opt}} = n/2$. Hence, the time complexity of $\mathcal{M}_{r,t}$ in PLPA⁺ is also $O(n \cdot m + d \cdot m)$.

Consequently, the time complexity for both PLPD⁺ and PLPA⁺ is $O(n \cdot m + d \cdot m)$. $\square$

8.6 Proofs of Theorems for Privacy Analysis

According to relationship between the definitions of (w, ϵ) -EPLDP, ϵ -PLDP, we have the following important three lemmas for privacy analysis.

Lemma 8.1. *Given any window size $w \in \mathbb{N}^+$, a mechanism $\mathcal{M}$ satisfies $(w \cdot 1, \epsilon)$ -EPLDP if it satisfies ϵ -PLDP in any window of size w .*

PROOF. Let S_t and $\hat{S}_t$ be any w -neighboring stream prefixes. If $S_t = \hat{S}_t$, then for any user $u_i \in \mathcal{U}$, it is obvious that $\Pr[\mathcal{M}(S_t)] \leq e^{\epsilon(u_i)} \cdot \Pr[\mathcal{M}(\hat{S}_t)]$. Otherwise, we can find $\tau_1, \tau_2 \in [t]$ satisfying $\tau_2 - \tau_1 + 1 \leq w$, with $S_t[\tau_1] \neq \hat{S}_t[\tau_1]$ and $S_t[\tau_2] \neq \hat{S}_t[\tau_2]$, meanwhile for any $\tau \in [1, \tau_1 - 1] \cup [\tau_2 + 1, t]$ it satisfies $S_t[\tau] = \hat{S}_t[\tau]$, and for any $\tau \in [\tau_1, \tau_2]$, it satisfies $S_t[\tau]$ and $\hat{S}_t[\tau]$ are neighboring.

Let $\tilde{\epsilon} = \text{Unique}(\epsilon)$ be the unique ϵ among all users. Let m be the number of $\tilde{\epsilon}$. For any two w length sub-stream segments $S_{\tau-w+1, \tau}$ and $\hat{S}_{\tau-w+1, \tau}$ satisfying $\tau - w + 1 \leq \tau_1 \leq \tau_2 \leq \tau$, since $\mathcal{M}$ satisfies ϵ -PLDP within any w window, then for any personalized sub-stream segment $S_{\tau-w+1, \tau}^{(k)} \subseteq S_{\tau-w+1, \tau}$ and $\hat{S}_{\tau-w+1, \tau}^{(k)} \subseteq \hat{S}_{\tau-w+1, \tau}$ we have $\forall y \in \text{Range}(\mathcal{M})$,

$$\Pr[\mathcal{M}(S_{\tau-w+1, \tau}^{(k)}) = y] \leq e^{\epsilon(u_i)} \Pr[\mathcal{M}(\hat{S}_{\tau-w+1, \tau}^{(k)}) = y].$$

Therefore, for any personalized stream prefix $S_t^{(k)} \subseteq S_t, \hat{S}_t^{(k)} \subseteq \hat{S}_t$ and $\forall y \in \text{Range}(\mathcal{M})$ with $y = y_{1:\tau-w} \| y_{\tau-w+1, \tau} \| y_{\tau, t}$, we have

$$\begin{aligned} & \frac{\Pr[\mathcal{M}(S_t^{(k)}) = y]}{\Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y]} \\ & \leq \frac{\Pr[\mathcal{M}(S_{1, \tau-w}^{(k)}) = y_{1, \tau-w}]}{\Pr[\mathcal{M}(\hat{S}_{1, \tau-w}^{(k)}) = y_{1, \tau-w}]} \times \frac{\Pr[\mathcal{M}(S_{\tau-w+1, \tau}^{(k)}) = y_{\tau-w+1, \tau}]}{\Pr[\mathcal{M}(\hat{S}_{\tau-w+1, \tau}^{(k)}) = y_{\tau-w+1, \tau}]} \\ & \quad \times \frac{\Pr[\mathcal{M}(S_{\tau+1, t}^{(k)}) = y_{\tau+1, t}]}{\Pr[\mathcal{M}(\hat{S}_{\tau+1, t}^{(k)}) = y_{\tau+1, t}]} \\ & = \frac{\Pr[\mathcal{M}(S_{\tau-w+1, \tau}^{(k)}) = y_{\tau-w+1, \tau}]}{\Pr[\mathcal{M}(\hat{S}_{\tau-w+1, \tau}^{(k)}) = y_{\tau-w+1, \tau}]} \\ & \leq e^{\epsilon(u_i)}. \end{aligned}$$

Therefore, $\mathcal{M}$ satisfies $(w \cdot 1, \epsilon)$ -EPLDP. $\square$

Lemma 8.2. *A mechanism $\mathcal{M}$ satisfies (w, ϵ) -EPLDP if it satisfies $(\max(w) \cdot 1, \epsilon)$ -EPLDP.*

PROOF. For any user u_i with $\epsilon(u_i) = \epsilon_k$ and $w(u_i) = w_k$, let $S_t^{(k)}, \hat{S}_t^{(k)} \subseteq S_t$ be any pair of w_k -neighboring stream prefixes. If $S_t^{(k)} = \hat{S}_t^{(k)}$, then it is evident that $\forall y \in \text{Range}(\mathcal{M}), \Pr[\mathcal{M}(S_t^{(k)}) = y] \leq e^{\epsilon_k} \cdot \Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y]$. Otherwise, there exists $\tau_1, \tau_2 \in [t]$ satisfying $\tau_2 - \tau_1 + 1 \leq w_k$ with $S_t[\tau_1] \neq \hat{S}_t[\tau_1]$ and $S_t[\tau_2] \neq \hat{S}_t[\tau_2]$. Because $w_k \leq \max(w)$, then we can choose $\tau'_1 \in [1, \tau_1], \tau'_2 \in [\tau_2, t]$ with $\tau'_2 - \tau'_1 + 1 = \max(w)$. Thus $S_t^{(k)} \sim_{\max(w)} \hat{S}_t^{(k)}$. Because $\mathcal{M}$ satisfies $(\max(w) \cdot 1, \epsilon)$ -EPLDP, then $\forall y \in \text{Range}(\mathcal{M})$, we have

$$\Pr[\mathcal{M}(S_t^{(k)}) = y] \leq e^{\epsilon_k} \cdot \Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y].$$

Therefore, $\mathcal{M}$ satisfies (w, ϵ) -EPLDP. $\square$

Lemma 8.3. *A mechanism $\mathcal{M}$ satisfies (w, ϵ) -EPLDP if for any window size $\hat{w} \in [\min(w), \max(w)]$, $\mathcal{M}$ satisfies $(\hat{w} \cdot 1, \hat{\epsilon})$ -EPLDP where*

$$\hat{\epsilon}(u_i) = \begin{cases} \epsilon(u_i), & \text{if } w(u_i) \leq \hat{w}, \\ \epsilon(u_i) / \lceil w(u_i) / \hat{w} \rceil, & \text{otherwise.} \end{cases}$$

PROOF. All users can be classified into two cases: (1) the users whose window sizes are no more than $\hat{w}$; (2) the users whose window sizes are larger than $\hat{w}$.

For any user u_i in case (1), it satisfies $w(u_i) = w_k \leq \hat{w}$. Then $\forall S_t^{(k)}, \hat{S}_t^{(k)}$ with $S_t^{(k)} \sim_{w_k} \hat{S}_t^{(k)}$, we have $S_t^{(k)} \sim_{\hat{w}} \hat{S}_t^{(k)}$. Because $\mathcal{M}$ satisfies $(\hat{w} \cdot 1, \hat{\epsilon})$ -EPLDP, we have $\forall y \in \text{Range}(\mathcal{M})$,

$$\frac{\Pr[\mathcal{M}(S_t^{(k)}) = y]}{\Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y]} \leq e^{\hat{\epsilon}(u_i)} = e^{\epsilon(u_i)}.$$

For any user u_i in case (2), it satisfies $w(u_i) = w_k > \hat{w}$. Let $S_t^{(k)}, \hat{S}_t^{(k)}$ be any two personalized stream prefixes satisfying $S_t \sim_{w_k} \hat{S}_t$. If $S_t^{(k)} = \hat{S}_t^{(k)}$, then it is obvious that $\Pr[\mathcal{M}(S_t^{(k)}) = y] /$

$\Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y] = 1 \leq e^{\epsilon(u_i)}$. Otherwise, we can find τ_1, τ_2 satisfying $\tau_2 - \tau_1 + 1 \leq w_k, S_t^{(k)}[\tau_1] \neq \hat{S}_t^{(k)}[\tau_1], S_t^{(k)}[\tau_2] \neq \hat{S}_t^{(k)}[\tau_2]$ and $\forall \tau \in [1, \tau_1 - 1] \cup [\tau_2 + 1, t], S_t^{(k)}[\tau] = \hat{S}_t^{(k)}[\tau]$. Let $z = \lceil w_k / \hat{w} \rceil$, then we can construct a list of $z - 1$ stream prefixes $\check{S} = \langle \check{S}_t^{(k,1)}, \check{S}_t^{(k,2)}, \dots, \check{S}_t^{(k,z-1)} \rangle$ satisfying $S_t^{(k)} \sim_{\hat{w}} \check{S}_t^{(k,1)}, \check{S}_t^{(k,1)} \sim_{\hat{w}} \check{S}_t^{(k,2)}$ and $\forall j \in [2, z - 1], \check{S}_t^{(k,j-1)} \sim_{\hat{w}} \check{S}_t^{(k,j)}$. By adding $S_t^{(k)}$ to the head of $\check{S}$ and $\hat{S}_t^{(k)}$ to tail, we can get the extended list of $z + 1$ stream prefixes $\check{S}^+ = \langle S_t^{(k)}, \check{S}_t^{(k,1)}, \check{S}_t^{(k,2)}, \dots, \check{S}_t^{(k,z-1)}, \hat{S}_t^{(k)} \rangle$. For any neighboring pair prefixes x, x' in $\check{S}^+$, because $\mathcal{M}$ satisfies $(\hat{w} \cdot 1, \epsilon)$ -EPLDP, we have $\forall y \in \text{Range}(\mathcal{M}), \Pr[\mathcal{M}(x) = y] / \Pr[\mathcal{M}(x') = y] \leq e^{\epsilon k / z}$. Therefore, for $\forall y \in \text{Range}(\mathcal{M})$, we have

$$\begin{aligned} & \frac{\Pr[\mathcal{M}(S_t^{(k)}) = y]}{\Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y]} \\ &= \frac{\Pr[\mathcal{M}(S_t^{(k)}) = y]}{\Pr[\mathcal{M}(\check{S}_t^{(k,1)}) = y]} \times \prod_{j=2}^{z-1} \frac{\Pr[\mathcal{M}(\check{S}_t^{(k,j-1)}) = y]}{\Pr[\mathcal{M}(\check{S}_t^{(k,j)}) = y]} \\ & \times \frac{\Pr[\mathcal{M}(\check{S}_t^{(k,z-1)}) = y]}{\Pr[\mathcal{M}(\hat{S}_t^{(k)}) = y]} \\ &\leq e^{\epsilon k / z} \times \left(e^{\epsilon k / z}\right)^{z-2} \times e^{\epsilon k / z} \\ &= e^{\epsilon k} \\ &= e^{\epsilon(u_i)}. \end{aligned}$$

Therefore, $\mathcal{M}$ satisfies (w, ϵ) -EPLDP. $\square$

Next, we give one special lemmas for the privacy analysis of PLPD and PLPA, and one corollary for that of PLPD⁺ and PLPA⁺.

Lemma 8.4. *For both PLPD and PLPA, in any time window composed of $w_{\max} = \max(w)$ consecutive time slots, each user reports to the server at most once.*

PROOF. We only need to prove that the number of users within any $w_{\max}$ window is no more than n , i.e., $\sum_{\tau=t-w_{\max}+1}^t |U_{\tau}| \leq n$. Then, because we sample a fresh set of users U_{τ} at each time slot τ satisfying $U_{s,\tau} \cap U_{r,\tau} = \emptyset$, we can guarantee each user reports only once within any window of size $w_{\max}$.

For PLPD, in $M_{s,t}$, $\lfloor n / (2w_{\max}) \rfloor$ users are allocated at each time slot t . Thus, for any t and $k \in [t]$, there are $\sum_{\tau=k-w_{\max}+1}^k |U_{s,\tau}| \leq n/2$ users. In $M_{r,t}$, it either publishes with additional users $U_{r,t}$ or approximates the last release without any user assignment. In the former case, there are at most $|U_{r,t}| = (n/2 - \sum_{\tau=k-w_{\max}+1}^{k-1} |U_{r,\tau}|) / 2$ users. Because $M_{r,t}$ always uses at most half of the available users, we have $0 \leq \sum_{\tau=k-w_{\max}+1}^k |U_{r,\tau}| \leq n/2$. In the latter case, $|U_{r,t}| = 0$. Therefore, for any t and $k \in [t]$, the total number of publication users within a $w_{\max}$ window for PLPD is

$$\sum_{\tau=k-w_{\max}+1}^k |U_{\tau}| = \sum_{\tau=k-w_{\max}+1}^k |U_{s,\tau}| + \sum_{\tau=k-w_{\max}+1}^k |U_{r,\tau}| \leq n.$$

For PLPA, similarly, in $M_{s,t}$, there are $\sum_{\tau=k-w_{\max}+1}^k |U_{s,\tau}| \leq n/2$ users for any t and $\tau \in [t]$. In $M_{r,t}$, assume there are at most c new publications in any $w_{\max}$ window. We denote these publication time

slots as $(\tau_1, \tau_2, \dots, \tau_c)$. For any publication time slot τ_j ($j \in [c]$), we denote the number of its absorbed population share as η_j , thus after absorption, time slot τ_j will occupy $\eta_j + 1$ population shares. Figure 9 illustrates an example for $c = 3$ and $w_{\max} = 9$, in which, for instance, time slot 3 absorbs population shares from time slots 1 and 2, while nullifying the population at time slots 4 and 5. Noted

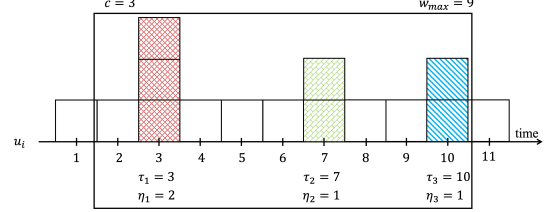


Figure 9: An example for new publications' absorption in PLPA.

that the absorbed time slot length and nullified time slot length are symmetric with respect to the new publication, according to the definition of τ_j , we have:

$$w_{\max} \geq \sum_{j=1}^c (1 + 2\eta_j) - \eta_1 - \eta_c.$$

Then, for any t and $k \in [t]$, we have

$$\begin{aligned} \sum_{\tau=k-w_{\max}+1}^k |U_{r,\tau}| &\leq \left\lfloor \frac{n}{2w_{\max}} \right\rfloor \cdot \sum_{j=1}^c (1 + \eta_j) \\ &\leq \frac{n \cdot \sum_{j=1}^c (1 + \eta_j)}{2 \sum_{j=1}^c (1 + 2\eta_j) - 2\eta_1 - 2\eta_c} \\ &= \frac{n \cdot \sum_{j=1}^c (1 + \eta_j)}{2 \sum_{j=1}^c (1 + \eta_j) + 2 \sum_{j=2}^{c-1} \eta_j} \\ &\leq n/2. \end{aligned}$$

Therefore, for any t and $k \in [t]$, the total number of publication users within a $w_{\max}$ window for PLPA is

$$\sum_{\tau=k-w_{\max}+1}^k |U_{\tau}| = \sum_{\tau=k-w_{\max}+1}^k |U_{s,\tau}| + \sum_{\tau=k-w_{\max}+1}^k |U_{r,\tau}| \leq n.$$

$\square$

From Lemma 8.4, we can get the following obvious corollary.

Corollary 8.1. *For both PLPD⁺ and PLPA⁺, in any time window composed of w_{opt} consecutive time slots, each user reports to the server at most once.*

8.6.1 Proof of Theorem 4.2.

PROOF. According to Lemma 8.4, for any user u_i in any window of size $w_{\max}$, it reports at most once. Thus for any two personalized $w_{\max}$ -neighboring stream segments $S_{\tau-w_{\max}+1,\tau}^{(k)}, \hat{S}_{\tau-w_{\max}+1,\tau}^{(k)}$, we have $\forall y \in \text{Range}(\mathcal{M})$,

$$\frac{\Pr[\mathcal{M}(S_{\tau-w_{\max}+1,\tau}^{(k)}) = y]}{\Pr[\mathcal{M}(\hat{S}_{\tau-w_{\max}+1,\tau}^{(k)}) = y]} \leq e^{\epsilon(u_i)}.$$

Hence, $\mathcal{M}$ satisfies ϵ -PLDP in any window of size $w_{\max}$. According to Lemma 8.1, $\mathcal{M}$ satisfies $(w_{\max} \cdot 1, \epsilon)$ -EPLDP. Then, according to Lemma 8.2, $\mathcal{M}$ satisfies (w, ϵ) -EPLDP. $\square$

8.6.2 *Proof of Theorems 5.3.* According to Corollary 8.1 and the process of OPS, we can conclude that both PLPD⁺ and PLPA⁺ $\mathcal{M}$ satisfy $\hat{\epsilon}$ -PLDP in any window of size w_{opt} , where $\hat{\epsilon}$ satisfies:

$$\hat{\epsilon}(u_i) = \begin{cases} \epsilon(u_i), & \text{if } w(u_i) \leq \hat{w}, \\ \epsilon(u_i) / \lceil w(u_i) / \hat{w} \rceil, & \text{otherwise.} \end{cases}$$

According to Lemma 8.1, $\mathcal{M}$ satisfies $(w_{opt} \cdot 1, \epsilon)$ -EPLDP. Then according to Lemma 8.3, we can conclude that PLPD⁺ and PLPA⁺ satisfy (w, ϵ) -EPLDP.

8.7 Proofs of Theorems for Utility Analysis

Lemma 8.5. *Given a series privacy budget-count set $P = \{(\epsilon_k, n_k) \mid \epsilon_k \in \tilde{\epsilon}, n_k \in \tilde{n}, \sum_k n_k = n\}$ with sampling size z from n users, where $\tilde{\epsilon}$ is the total distinct privacy budget set with $\forall \epsilon_i, \epsilon_j \in \tilde{\epsilon}, \epsilon_i \neq \epsilon_j$ and $\tilde{n}$ is the all distinct appearing count. (ϵ_k, n_k) indicates that there are n_k users proposing privacy budget requirement ϵ_k . The error upper bound of FO process is :*

$$\widetilde{err}_{FO}(P, z) = \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}. \quad (16)$$

Let $\hat{n}_{\max}(\tilde{\epsilon})$ be the user number of requiring the maximum privacy budget $\max(\tilde{\epsilon})$, the error upper bound of PFO process is :

$$\widetilde{err}_{PFO}(P, z) = \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \hat{n}_{\max}(\tilde{\epsilon})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}. \quad (17)$$

PROOF. For the FO case, according to Equation (6) by setting $p_k = \frac{e^{\epsilon_k}}{e^{\epsilon_k} + d - 1}$, $q_k = \frac{1}{e^{\epsilon_k} + d - 1}$, and denoting $\lambda_k = \frac{q_k(1-q_k)}{g_k(p_k - q_k)^2}$, $\mu_k = \frac{1-p_k-q_k}{g_k(p_k - q_k)}$, we can get the error as

$$\begin{aligned} err_{FO}(P, z) &= \frac{n-z}{z(n-1)} + \frac{1/z}{\sum_{k=1}^m 1/(d\lambda_k + \mu_k)} \\ &\leq \frac{n-z}{z(n-1)} + \frac{1}{zm} \sum_{k=1}^m (d\lambda_k + \mu_k) \\ &= \frac{n-z}{z(n-1)} + \frac{d-1}{zm} \sum_{k=1}^m \frac{1}{g_k} \left(\frac{2}{e^{\epsilon_k} - 1} + \frac{d}{(e^{\epsilon_k} - 1)^2} \right) \\ &= \frac{n-z}{z(n-1)} + \frac{d-1}{zm} \sum_{k=1}^m \frac{n}{n_k} \left(\frac{2}{e^{\epsilon_k} - 1} + \frac{d}{(e^{\epsilon_k} - 1)^2} \right) \\ &\leq \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \min(\tilde{n})} \cdot \left(\frac{2}{e^{\min(\tilde{\epsilon})} - 1} + \frac{d}{(e^{\min(\tilde{\epsilon})} - 1)^2} \right) \\ &= \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}. \end{aligned} \quad (18)$$

For the PFO case, without loss generation, let $\min(\tilde{\epsilon}) = \epsilon_1 < \epsilon_2 < \dots < \epsilon_m = \max(\tilde{\epsilon})$. According to Theorem 5.1, for users with privacy budget ϵ_k , the error decreases from $\frac{1}{n_k} \cdot \left(\frac{dq_k(1-q_k)}{(p_k - q_k)^2} + \frac{1-p_k-q_k}{p_k - q_k} \right) = \frac{g_k}{n_k} \cdot (d\lambda_k + \mu_k)$ to $\frac{1}{\sum_{j=k}^m n_j} \cdot \left(\frac{dq_k(1-q_k)}{(p_k - q_k)^2} + \frac{1-p_k-q_k}{p_k - q_k} \right) = \frac{g_k}{\sum_{j=k}^m n_j} \cdot (d\lambda_k + \mu_k)$. Namely, λ_k varies to $\lambda'_k = \frac{n_k}{\sum_{j=k}^m n_j} \cdot \lambda_k$ and μ_k varies to $\mu'_k = \frac{n_k}{\sum_{j=k}^m n_j} \cdot \mu_k$.

For the error $err_{PFO}(P, z)$, we have

$$\begin{aligned} err_{PFO}(P, z) &= \frac{n-z}{z(n-1)} + \frac{1/z}{\sum_{k=1}^m 1/(d\lambda'_k + \mu'_k)} \\ &\leq \frac{n-z}{z(n-1)} + \frac{1}{zm} \sum_{k=1}^m (d\lambda'_k + \mu'_k) \\ &= \frac{n-z}{z(n-1)} + \frac{d-1}{zm} \sum_{k=1}^m \frac{n}{\sum_{j=k}^m n_j} \left(\frac{2}{e^{\epsilon_k} - 1} + \frac{d}{(e^{\epsilon_k} - 1)^2} \right) \\ &\leq \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \hat{n}_{\max}(\tilde{\epsilon})} \cdot \left(\frac{2}{e^{\min(\tilde{\epsilon})} - 1} + \frac{d}{(e^{\min(\tilde{\epsilon})} - 1)^2} \right) \\ &= \frac{n-z}{z(n-1)} + \frac{n(d-1)}{z \hat{n}_{\max}(\tilde{\epsilon})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}. \end{aligned}$$

□

8.7.1 Proof of Theorem 4.3 and 4.4.

PROOF. Let $P = \{(\epsilon_k, n_k) \mid \epsilon_k \in \tilde{\epsilon}, n_k \in \tilde{n}, \sum_k n_k = n\}$ be the distinct privacy budget-count set. At any time slot t , if the sampling size is z , according to Equation (16) in Lemma 8.5, the error upper bound of executing FO is $\frac{n}{z(n-1)} - \frac{1}{n-1} + \frac{n(d-1)}{z \min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}$.

Suppose $\mathcal{M}_{s,t}$ is not private, then the error only comes from sub-mechanism $\mathcal{M}_{r,t}$. We discuss $\mathcal{M}_{r,t}$ in PLPD and PLPA, respectively.

In PLPD, for any new publication at time slot τ , the error is $\widetilde{err}_{FO}(P, z)$, where z is the sampling size at τ . For other skipped time slots, the error is no larger than that of the last new publication. Since the population is distributed to the s new publications in an exponentially decreasing way, the sampling population size sequence is then $n/4, n/8, \dots, n/2^{s+1}$. Thus, by denoting $A = \frac{d-1}{\min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}$ we have

$$\begin{aligned} \widetilde{err}_{PLPD,r,t} &= \frac{1}{w_{\max}} \cdot \sum_{\tau=1}^s \frac{w_{\max}}{s} \cdot err_{FO}\left(P, \frac{n}{2^{\tau+1}}\right) \\ &\leq \frac{1}{s} \cdot \sum_{\tau=1}^s \widetilde{err}_{FO}\left(P, \frac{n}{2^{\tau+1}}\right) \\ &= \frac{1}{s} \sum_{\tau=1}^s \left(\frac{n}{2^{\tau+1}(n-1)} - \frac{1}{n-1} + \frac{n}{2^{\tau+1}} \cdot A \right) \\ &= \frac{1}{s} \cdot \left(-\frac{s}{n-1} + \left(\frac{1}{n-1} + A \right) \sum_{\tau=1}^s 2^{\tau+1} \right) \\ &= \frac{1}{s} \cdot \left(-\frac{s}{n-1} + \left(\frac{1}{n-1} + A \right) \cdot 4 \cdot (2^s - 1) \right) \\ &= \frac{4(2^s - 1)}{s} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}. \end{aligned} \quad (19)$$

In PLPA, since there are s new publications, there are $w_{\max} - s$ approximate publications. Because the skipped and nullified publications occur in pairs (i.e., the numbers of skipped publications and nullified publications are equal) and each new publication corresponds the same number of skipped publications, the numbers of skipped and nullified publications are both $n_{skp} = n_{nlf} = \frac{w_{\max} - s}{2 \cdot s}$. For each new publication, the population size is $n_{skp} \cdot \frac{n}{2w_{\max}} = \frac{(w_{\max} - s)n}{4s w_{\max}}$. For each skipped publication, its error is no more than the new publication, otherwise it will not skip. Further more, a new publication

sampling size lower bound at each skipped publication time slot is increasing from $\frac{n}{2w_{\max}}$ in an exponential fashion, namely, they are $\frac{n}{2w_{\max}}, \frac{n \cdot 2}{2w_{\max}}, \dots, \frac{n \cdot 2^{n_{\text{skp}}}}{2w_{\max}}$. For each nullified publication, because we have no information about the underlying statistic and nullification is enforced prior to their arrival, we cannot quantify this kind of error. Thus, we denote the nullified publication error as err_{nlf} . For the error of $\mathcal{M}_{r,t}$ in PLPA, by denoting $A = \frac{d-1}{\min(\tilde{n})} \cdot \frac{2e^{\min(\tilde{\epsilon})} + d - 2}{(e^{\min(\tilde{\epsilon})} - 1)^2}$, $B = n_{\text{skp}} = \frac{w_{\max} - s}{2s}$ and $C = \frac{s}{w_{\max}}$, we have

$$\begin{aligned} \overline{\text{err}}_{\text{PLPA}^+, r, t} &\leq \frac{1}{w_{\max}} \cdot \sum_{\tau=1}^s \left(\widetilde{\text{err}}_{\text{FO}} \left(P, \frac{(w_{\max} - s)n}{4s w_{\max}} \right) + \sum_{j=1}^{2s} \widetilde{\text{err}}_{\text{FO}} \left(P, \frac{n \cdot 2^j}{2w_{\max}} \right) \right. \\ &\quad \left. + \frac{w_{\max} - s}{2s} \cdot \text{err}_{\text{nlf}} \right) \\ &= \frac{s}{w_{\max}} \cdot \left(-\frac{1}{n-1} + \frac{4s w_{\max}}{(n-1) \cdot (w_{\max} - s)} + \frac{4s w_{\max}}{(w_{\max} - s)} \cdot A - \frac{w_{\max} - s}{2s(n-1)} \right. \\ &\quad \left. + \left(\frac{w_{\max}}{n-1} + w_{\max} \cdot A \right) \cdot \sum_{j=1}^{2s} \frac{1}{2^{j-1}} + \frac{w_{\max} - s}{2s} \cdot \text{err}_{\text{nlf}} \right) \\ &= C \cdot \left(B \cdot \text{err}_{\text{nlf}} - \frac{B+1}{n-1} + \left(\frac{w_{\max}}{n-1} + w_{\max} \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right) \right) \\ &= BC \cdot \text{err}_{\text{nlf}} - \frac{(B+1)C}{n-1} + \left(\frac{s}{n-1} + s \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right). \end{aligned} \quad (20)$$

When sub-mechanism $\mathcal{M}_{s,t}$ is private, in both PLPD and PLPA, there is a sample $|U_{s,t}|$ of fixed sampling size $z_{\min} = \lfloor n/(2w_{\max}) \rfloor$ of users for dis calculation. The error in $\mathcal{M}_{s,t}$ will lead to a *false* publication decision, i.e., making $\mathcal{M}_{r,t}$ (1) falsely skip a publication, or (2) falsely perform a new publication. Both of the two cases are due to the error of dis calculation, which is $\frac{1}{d^3} \text{err}_{\text{FO}}(P, z_{\min})^2$. Therefore the average error of $\mathcal{M}_{s,t}$ within any window of size $w_{\max}$ is

$$\begin{aligned} \overline{\text{err}}_{s,t} &\leq \frac{1}{d^3} \cdot \widetilde{\text{err}}_{\text{FO}}(P, z_{\min})^2 \\ &= \frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\max}}{n-1} + 2w_{\max}A \right)^2. \end{aligned} \quad (21)$$

Thus, the error upper bound at each time slot is $\frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\max}}{n-1} + 2w_{\max}A \right)^2 + \frac{4(2^s-1)}{s} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}$ in PLPD and $\frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\max}}{n-1} + 2w_{\max}A \right)^2 + BC \cdot \text{err}_{\text{nlf}} - \frac{(B+1)C}{n-1} + \left(\frac{s}{n-1} + s \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right)$ in PLPA. $\square$

8.7.2 Proof of Theorem 5.4 and 5.5.

PROOF. Let $P_{\text{opt}} = \{(\epsilon_k, n_k) | \epsilon_k \in \tilde{\epsilon}_{\text{opt}}, n_k \in \tilde{n}_{\text{opt}}, \sum_k n_k = n\}$ be the distinct privacy budget-count set after executing OPS. At any time slot t , if the sampling size is z , according to Equation (17) in Lemma 8.5, the error upper bound of executing PFO is $\frac{n}{z(n-1)} - \frac{1}{n-1} + \frac{n(d-1)}{z\tilde{n}_{\max}(\tilde{\epsilon}_{\text{opt}})} \cdot \frac{2e^{\min(\tilde{\epsilon}_{\text{opt}})} + d - 2}{(e^{\min(\tilde{\epsilon}_{\text{opt}})} - 1)^2}$.

Similar to the process in utility process (Section 8.7.1) of PLPD and PLPA. We first assume the sub-mechanism $\mathcal{M}_{s,t}$ is not private.

In this way, for the error of $\mathcal{M}_{r,t}$ in PLPD⁺, by setting $A = \frac{d-1}{\tilde{n}_{\max}(\tilde{\epsilon}_{\text{opt}})} \cdot \frac{2e^{\min(\tilde{\epsilon}_{\text{opt}})} + d - 2}{(e^{\min(\tilde{\epsilon}_{\text{opt}})} - 1)^2}$, we have

$$\begin{aligned} \overline{\text{err}}_{\text{PLPD}^+, r, t} &= \frac{1}{w_{\text{opt}}} \cdot \sum_{\tau=1}^{s_{\text{opt}}} \frac{w_{\text{opt}}}{s_{\text{opt}}} \cdot \text{err}_{\text{PFO}} \left(P_{\text{opt}}, \frac{n}{2^{\tau+1}} \right) \\ &\leq \frac{4(2^{s_{\text{opt}}} - 1)}{s_{\text{opt}}} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}. \end{aligned}$$

For the error of $\mathcal{M}_{r,t}$ in PLPD⁺, by further setting $B = \frac{w_{\text{opt}} - s_{\text{opt}}}{2s_{\text{opt}}}$ and $C = \frac{s_{\text{opt}}}{w_{\text{opt}}}$, we have

$$\begin{aligned} \overline{\text{err}}_{\text{PLPA}^+, r, t} &\leq \frac{1}{w_{\text{opt}}} \cdot \sum_{\tau=1}^{s_{\text{opt}}} \left(\widetilde{\text{err}}_{\text{PFO}} \left(P_{\text{opt}}, \frac{(w_{\text{opt}} - s_{\text{opt}})n}{4s w_{\text{opt}}} \right) + \sum_{j=1}^{2s_{\text{opt}}} \widetilde{\text{err}}_{\text{PFO}} \left(P, \frac{n \cdot 2^j}{2w_{\text{opt}}} \right) \right. \\ &\quad \left. + \frac{w_{\text{opt}} - s_{\text{opt}}}{2s_{\text{opt}}} \cdot \text{err}_{\text{nlf}} \right) \\ &= BC \cdot \text{err}_{\text{nlf}} - \frac{(B+1)C}{n-1} + \left(\frac{s_{\text{opt}}}{n-1} + s_{\text{opt}} \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right). \end{aligned}$$

When $\mathcal{M}_{s,t}$ is private, we can get the error as

$$\begin{aligned} \overline{\text{err}}_{s,t} &\leq \frac{1}{d^3} \cdot \widetilde{\text{err}}_{\text{PFO}}(P, z_{\text{opt}})^2 \\ &= \frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\text{opt}}}{n-1} + 2w_{\text{opt}}A \right)^2. \end{aligned}$$

Therefore, the error upper bound is $\frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\text{opt}}}{n-1} + 2w_{\text{opt}}A \right)^2 + \frac{4(2^{s_{\text{opt}}} - 1)}{s_{\text{opt}}} \cdot \left(\frac{1}{n-1} + A \right) - \frac{1}{n-1}$ in PLPD⁺, and $\frac{1}{d^3} \left(-\frac{1}{n-1} + \frac{2w_{\text{opt}}}{n-1} + 2w_{\text{opt}}A \right)^2 + BC \cdot \text{err}_{\text{nlf}} - \frac{(B+1)C}{n-1} + \left(\frac{s_{\text{opt}}}{n-1} + s_{\text{opt}} \cdot A \right) \cdot \left(2 - \frac{1}{2^{B+1}} + \frac{2}{B} \right)$ in PLPA⁺. $\square$

8.8 Extra Experiment Results

We further evaluate each method in terms of the *Average Jensen-Shannon Divergence* [24] ($A\mathcal{JSD}, \bar{D}_{JS}$), defined as:

$$\begin{aligned} \bar{D}_{JS}(\hat{f} \| f) &= \frac{1}{T} \sum_{\tau=1}^T D_{JS}(\hat{f} \| f) = \frac{1}{T} \sum_{\tau=1}^T \left(\frac{1}{2} D_{KL}(\hat{f} \| \mathbf{v}) + \frac{1}{2} D_{KL}(f \| \mathbf{v}) \right) \\ &= \frac{1}{2T} \sum_{\tau=1}^T \sum_{j=1}^d \left(\hat{f}_{j,\tau} \log \left(\frac{\hat{f}_{j,\tau}}{\mathbf{v}_{j,\tau}} \right) + f_{j,\tau} \log \left(\frac{f_{j,\tau}}{\mathbf{v}_{j,\tau}} \right) \right), \end{aligned}$$

where $\mathbf{v}_{j,\tau} = \frac{1}{2} (\hat{f}_{j,\tau} + f_{j,\tau})$.

The experimental results, shown in Figures 10 to 13, are consistent with those observed under the AMSE metric.

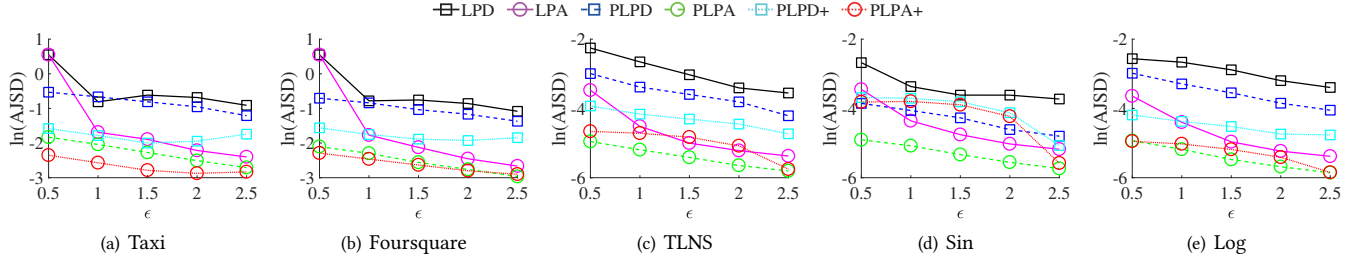


Figure 10: $AJSD$ with ϵ varied.

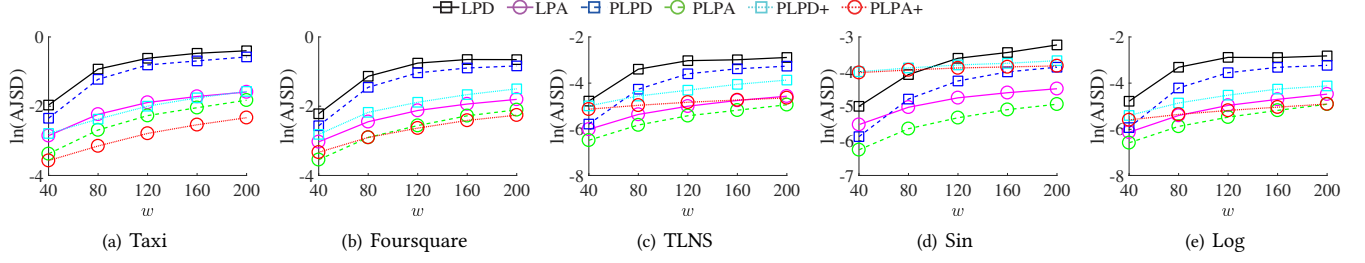


Figure 11: $AJSD$ with w varied.

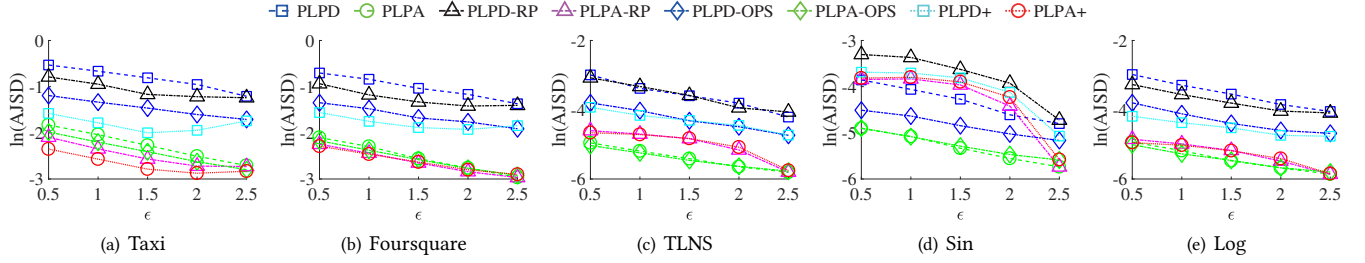


Figure 12: The ablation study of $AJSD$ with ϵ varied.

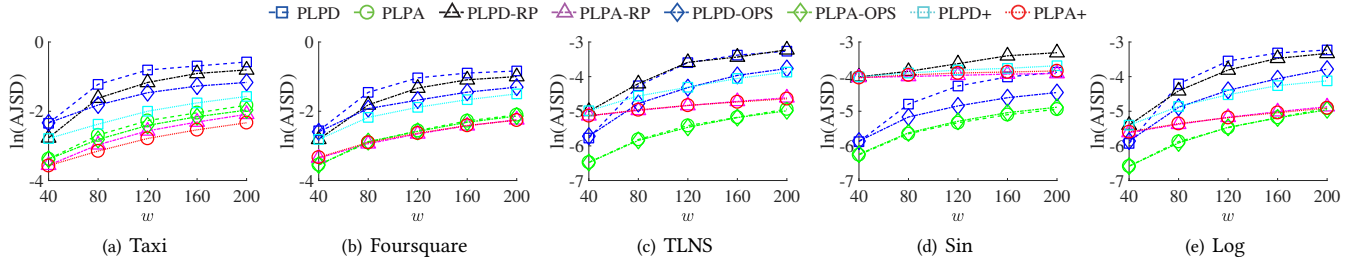


Figure 13: The ablation study of $AJSD$ with w varied.